Data Science Across Domains

From Methods and Models to Business Applications

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Welcome

Welcome to my portfolio:

Data Science Across Domains
From Methods and Models to Business Applications

This portfolio presents a curated collection of applied projects developed across different domains. Rather than a traditional theoretical text, it is organized as a series of case studies illustrating how quantitative methods, machine learning, and computational modeling can be used to address real-world problems.

Each chapter focuses on a specific project. Problems are clearly defined, modeling assumptions are made explicit, and solutions are developed through a combination of statistical reasoning, computational experimentation, and quantitative modeling. Results are analyzed and interpreted within the context of the problem being addressed.

The emphasis throughout the portfolio is on applied problem solving. Mathematical concepts and technical details are introduced only when they help clarify modeling decisions, support methodological choices, or explain the behavior of the proposed approaches.

Every project presented here corresponds to a fully implemented workflow. The associated computational artifacts—written in Python and R—are available alongside the text, allowing readers to explore the code, reproduce experiments, and examine the implementation details supporting each solution.

This portfolio reflects an approach to applied data science and quantitative modeling that combines structured reasoning, careful experimentation, and practical implementation.

Many of the projects presented here also draw on my background in physics and computational modeling, where mathematical reasoning and computational experimentation naturally complement data-driven approaches to problem solving.

Each chapter can be read independently as a standalone project, while together they illustrate a consistent approach to developing technically sound and practically meaningful solutions across different domains.

Preface

This portfolio reflects an ongoing effort to organize my work in applied data science and quantitative modeling into a coherent and practice-oriented body of projects. Rather than presenting isolated experiments or purely theoretical discussions, it documents complete workflows in which problem formulation, modeling choices, implementation, and evaluation are treated as parts of the same analytical process.

Structuring this material in book form mirrors how I approach quantitative and computational problems in professional and research environments. In practice, successful applications rarely depend on sophisticated methods alone. They usually emerge from a clear definition of the problem, explicit modeling assumptions, careful experimentation, and thoughtful interpretation of results within the context of the domain.

For this reason, the chapters are organized around concrete case studies. Each project begins with a well-defined problem and progresses through methodological design, implementation, evaluation, and discussion of limitations. Concepts from statistics, optimization, probability, numerical methods, and machine learning appear throughout the text, but only when they are necessary to support modeling decisions or clarify the behavior of the proposed approaches.

An equally important aspect of this portfolio is its deliberate scope. It does not attempt to catalogue every possible technique or follow trends for their own sake. Instead, it focuses on methods that I have implemented, tested, and evaluated in real problem settings. The emphasis is placed on depth, clarity, and practical relevance rather than on breadth.

All projects presented here correspond to fully implemented solutions. Computational artifacts written in Python and R accompany the text and are referenced throughout, allowing readers to explore the implementation and reproduce the results when appropriate.

Ultimately, this portfolio serves both as a record of completed work and as a foundation for continued development. The projects documented here represent practical steps in an ongoing process of learning, experimentation, and refinement. My intention is that the structure of this work reflects not only technical capability, but also a disciplined and pragmatic approach to quantitative modeling and computational problem solving in real-world contexts.

About

I am a physicist and data scientist with a background in applied mathematics, machine learning, and computational modeling. My work focuses on applying quantitative methods to real-world problems, combining mathematical reasoning with practical implementation.

My academic training in physics has strongly shaped the way I approach quantitative modeling and data-driven problem solving. My research included work on **quantum optimal control theory**, with particular emphasis on algorithmic design based on *shortcuts to adiabaticity* and the study of computational complexity in physical systems. This background continues to influence my modeling philosophy through an emphasis on clear problem formulation, explicit assumptions, and careful evaluation of model behavior.

Professionally, I have applied machine learning, statistical modeling, and optimization techniques across data-intensive industries. Much of my work has been developed within the **energy sector**, where I have contributed to quantitative modeling, decision-support systems, and optimization frameworks for gas trading and operational planning.

In the retail sector, I have worked on projects involving customer segmentation, churn prediction, and recommendation systems, balancing quantitative modeling with practical constraints such as data quality, interpretability, and deployment considerations.

More recently, within Oil & Gas research and development environments, my work has focused on integrating physical modeling, statistical inference, and machine learning to address complex industrial problems. These projects include predictive modeling for pipeline corrosion, production optimization, and simulation-based approaches for decision-making under uncertainty.

Across both academic and industrial settings, my perspective remains consistent: computational and statistical methods are most effective when supported by disciplined modeling, transparent assumptions, and careful validation.

Outside my professional work, I enjoy playing the bass guitar and maintain a longstanding interest in theology.

How to Read This Portfolio

This portfolio is organized around applied projects rather than purely theoretical chapters. Each chapter presents a self-contained case study illustrating how quantitative methods, machine learning, and computational modeling can be used to address real-world problems across different domains.

Projects can be read independently. Readers interested in a particular topic may navigate directly to the corresponding chapter without following the entire sequence.

Most projects follow a common structure:

1. **Problem Definition**
   Description of the problem, its context, and its practical relevance within the domain.

2. **Modeling Approach**
   Presentation of the conceptual framework, modeling assumptions, and methodological reasoning underlying the proposed solution.

3. **Methodology and Implementation**
   Description of the data, computational workflow, experimental design, and implementation details used throughout the project.

4. **Results, Discussion, and Limitations**
   Presentation and interpretation of results together with an analysis of model behavior, assumptions, limitations, and possible extensions.

5. **Technical Appendix (Optional)**
   Supplementary technical material, including mathematical derivations, methodological notes, and additional computational details when relevant to the project.

Theoretical concepts and mathematical details are introduced only when they help clarify modeling decisions, support methodological choices, or explain the behavior of the proposed approaches. The emphasis throughout the portfolio remains on applied problem solving, structured reasoning, and practical implementation rather than on purely theoretical discussion.

Each chapter is accompanied by computational artifacts implemented in `Python` and `R`, allowing readers to explore the implementation details, reproduce experiments, and examine complementary analyses when appropriate.

The material presented here combines quantitative reasoning, statistical modeling, machine learning methods, and computational experimentation, with a consistent focus on developing solutions that are both technically sound and practically meaningful.

Part I

Quantitative Finance

Overview

This part covers problems related to quantitative finance, including the modeling of financial assets, portfolio construction, and risk analysis under uncertainty.

The scope of this domain includes methods for analyzing market data, evaluating financial instruments, and supporting decision-making where risk and return must be jointly considered.

Chapter 1

CAPM and Fama–French Models

This project studies how classical and multi-factor asset pricing models explain the relationship between systematic risk and expected return. Using historical market data, the analysis evaluates the assumptions, explanatory power, and empirical limitations of the CAPM and Fama–French frameworks through regression-based factor modeling.

1.1 Problem Definition

This project evaluates how asset pricing models—from the traditional CAPM to the multi-factor extensions of Fama and French—formalise the relationship between risk and expected return. In practice, an investor faced with a price signal must determine whether the potential return justifies the associated risk. Doing this effectively means discounting uncertain future cash flows back to present value, which requires a reliable discount rate. The core question is straightforward: what minimum return should be demanded from an asset given its specific risk profile?

Getting this right is critical for asset valuation, portfolio construction, and fund performance evaluation. Without a structured framework, investment decisions often default to guesswork or oversimplified rules of thumb—such as chasing past returns or assuming that any form of risk automatically deserves a premium. Modern financial theory offers a more rigorous perspective: the market does not reward idiosyncratic or stock-specific risk, since it can be diversified away.

Only systematic risk warrants a higher expected return. This principle is what separates the CAPM from more naive approaches and serves as the foundation for multi-factor models.

Key decisions—ranging from efficient capital allocation to identifying market mispricings—rely heavily on these models. If an asset offers an expected return above what its systematic risk predicts, it may indicate undervaluation. However, the real-world challenge is methodological: true expected returns are unobservable. Relying on historical data introduces noise, sample-period sensitivity, and specification errors. Furthermore, the CAPM compresses all systematic risk into a single factor (the market portfolio), a simplification that empirical evidence has repeatedly challenged.

The Fama–French models attempt to address this limitation by capturing additional dimensions of systematic risk, including size, value, profitability, and investment patterns. From a data science perspective, this project explores how these models are constructed, tests their underlying assumptions, and evaluates how well they fit empirical data. The final output is a comparative analysis of the models' explanatory power and practical limitations when confronted with real-world financial data that rarely behaves as textbook theory predicts.

1.2 Modeling Approach

To approach the problem of asset valuation, this project is grounded on two fundamental principles of finance. The first is the time value of money: a dollar today is worth more than a dollar tomorrow because it can be invested immediately and start generating returns. The second is risk aversion: an uncertain cash flow is only attractive if it offers additional compensation relative to a risk-free alternative. Together, these principles reflect the two central variables underlying any financial valuation problem: time and risk.

Under this framework, the valuation of an asset can be understood as the present value of an uncertain future cash flow:

$$PV = \left(\frac{1}{1 + r_t}\right)^t \mathbb{E}[X_t] \tag{1.1}$$

Here, PV represents the expected present value of a future cash flow X_t received at time t, while r_t denotes the discount rate associated with risk and the opportunity cost of capital. The fundamental challenge for the investor is that this discount rate is not known in advance. In other words, the key question is how much return should be required to compensate for the level of risk being taken. This problem provides the conceptual motivation for the Capital Asset Pricing Model (CAPM) and its multifactor extensions.

Since future returns are uncertain, this project models them as random variables. A practical way to characterize them is through their expected value, representing the anticipated average return, and their volatility, which measures the dispersion of outcomes around that average. This mean-variance framework, originally developed by Harry Markowitz in modern portfolio theory, shows that total portfolio risk can be reduced by combining assets whose returns do not move exactly in the same way [1]. The intuition is straightforward: through diversification, part of the individual fluctuations of each asset tends to offset one another.

However, diversification has an observable practical limit. As portfolios are constructed with an increasing number of assets, total volatility initially falls rapidly but eventually stabilizes around a lower bound. As illustrated in Figure 1.1, this behavior reveals the existence of two distinct components of risk. On the one hand, idiosyncratic or diversifiable risk, which is associated with firm-specific events and can largely be eliminated through diversification. On the other hand, systematic risk, which is linked to broader macroeconomic and market-wide movements and therefore persists even in highly diversified portfolios.

This distinction has a fundamental implication for asset pricing. In a competitive market, rational investors should not expect compensation for taking risks that can be eliminated at virtually no cost through diversification. As a result, the market only rewards exposure to systematic risk. This idea forms the conceptual foundation of the CAPM, which states that the expected risk premium of an asset should be proportional to the expected risk premium of the market [2] [3]:

$$\mathbb{E}[r_i] - r_f = \beta_i \left(\mathbb{E}[r_m] - r_f \right) \tag{1.2}$$

In this expression, $\mathbb{E}[r_i]$ represents the expected return of asset i, r_f is the risk-free rate, and $\mathbb{E}[r_m]$ corresponds to the expected return of the market portfolio. The parameter β_i measures the sensitivity of the asset to market fluctuations and summarizes its exposure to systematic risk.

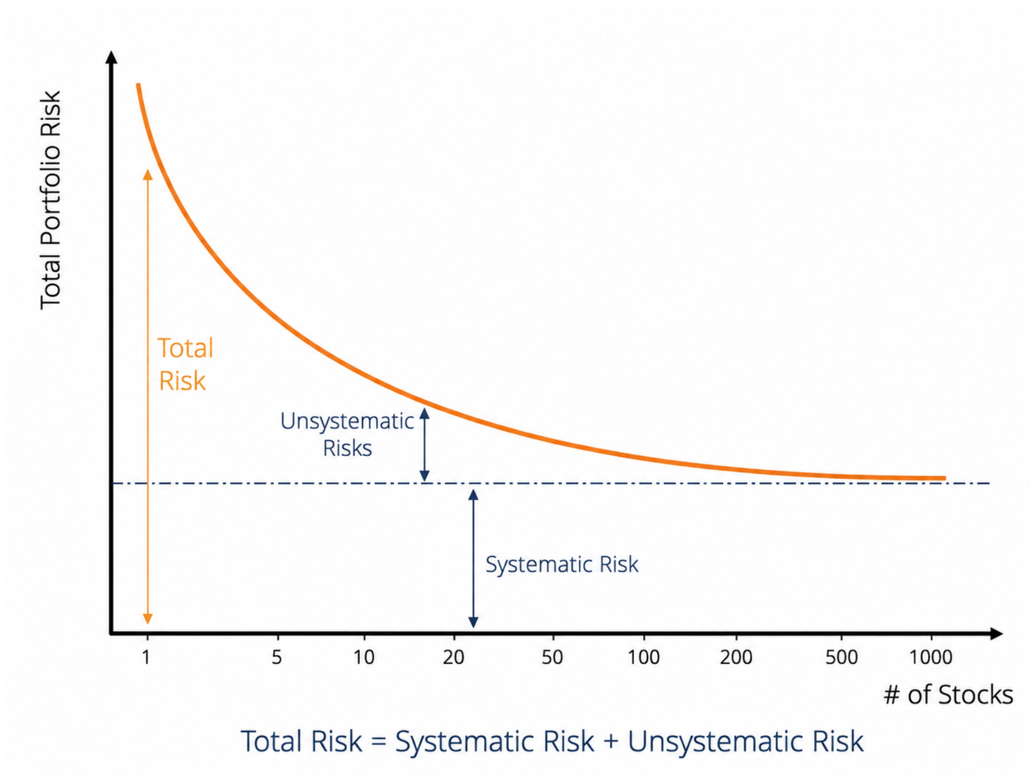

Figure 1.1: Decomposition of total risk into systematic and diversifiable risk as assets are progressively added to a portfolio.

From a modeling perspective, the estimation of beta can be interpreted as a simple linear regression between asset returns and market returns [4]. In this project, the CAPM is implemented precisely under this framework, using simple linear regression to estimate the market exposure of different assets and analyze the relationship between systematic risk and expected return.

Despite its conceptual simplicity and historical importance, the CAPM assumes that the entire structure of systematic risk can be summarized by a single factor: the market itself. Empirical evidence, however, suggests that certain firm characteristics — such as company size or the relationship between book value and market value — capture persistent return patterns that the CAPM does not fully explain. These observations motivated the development of the multifactor models proposed by Fama and French [5] [6].

In this project, these extensions are implemented using multiple linear regression. The three-factor model can be expressed as:

$$\mathbb{E}[r_i] - r_f = \alpha + \beta_{mkt}(\mathbb{E}[r_m] - r_f) + \beta_{smb} \cdot \text{SMB} + \beta_{hml} \cdot \text{HML} + \varepsilon \tag{1.3}$$

where SMB (*Small Minus Big*) measures the excess return of small-cap stocks relative to large-cap stocks, while HML (*High Minus Low*) captures the return differential between value and growth stocks [7] [8].

From a modeling perspective, the transition from the CAPM to the Fama–French three-factor model can be interpreted as moving from a simple linear regression to a multiple linear regression framework, incorporating additional explanatory variables to capture dimensions of systematic risk that a single-factor model cannot fully represent.

The main objective of the project is not simply to compare statistical goodness-of-fit metrics, but to understand how different economic factors contribute to the generation of returns in financial markets and to evaluate how effectively models of different complexity levels represent that underlying process.

1.3 Methodology and Implementation

1.3.1 Data Collection and Preprocessing

The analysis uses daily adjusted closing prices obtained from Yahoo Finance for both Apple Inc. (`AAPL`) and the S&P 500 index (`^GSPC`). The S&P 500 is used as a proxy for the market portfolio within the CAPM framework.

The selected period spans from January 2020 to December 2024, providing a multi-year sample that includes different market conditions and volatility regimes. Adjusted prices are used in order to account for corporate actions such as stock splits and dividend adjustments when available.

After downloading the data, both time series are merged into a single dataset and aligned by trading date. Missing observations are removed prior to return calculation.

The price series are initially inspected to verify data consistency and to obtain a preliminary view of the market dynamics during the analyzed period.

Daily simple returns are then computed from the adjusted closing prices. These returns constitute the main variables used throughout the regression analysis.

Before estimating the model, the relationship between asset and market returns is visually explored through a scatter plot. As shown in Figure 1.3, the data already suggest a positive linear relationship between Apple returns and market returns, supporting the use of a linear regression framework for the CAPM estimation.

1.3.2 CAPM Implementation

Following the theoretical formulation introduced in Equation 1.2, the CAPM is implemented through a simple linear regression in which the daily returns of Apple Inc. are modeled as a function of the daily returns of the market portfolio.

For simplicity, the implementation presented here uses raw daily returns without explicitly incorporating a risk-free rate. Under this approximation, the model estimates the linear sensitivity of the selected asset relative to market movements.

The estimated regression coefficients are summarized below.

Figure 1.2: Adjusted closing prices for Apple Inc. and the S&P 500 index between 2020 and 2024. The figure is mainly used as a preliminary inspection of the downloaded financial time series before return calculation.

Figure 1.3: Scatter plot of daily Apple returns against S&P 500 daily returns. The figure provides an exploratory visualization of the positive relationship between the asset and market returns prior to the CAPM estimation.

Table 1.1: Estimated CAPM regression coefficients for Apple Inc. using the S&P 500 as the market proxy.

Term	Estimate	Std. Error	t-statistic	p-value
(Intercept)	0.000526	0.000346	1.5226	0.128105
market	1.173257	0.025682	45.6847	0.000000

The estimated market beta is approximately $\beta = 1.173257$, indicating that Apple exhibited higher sensitivity to market fluctuations than the S&P 500 during the analyzed period. Under the CAPM interpretation, this coefficient suggests that, on average, Apple returns moved approximately 17% more aggressively than the market portfolio in response to market variations.

Additional model diagnostics are reported below.

Table 1.2: Diagnostic statistics associated with the CAPM linear regression model.

Metric	Value
R-squared	0.6247
Adjusted R-squared	0.6244
F-statistic	2087.0916
Observations	1256

The estimated beta coefficient is also statistically significant, with a large associated t-statistic and a near-zero p-value. Similarly, the overall regression presents a large F-statistic, supporting the existence of a statistically meaningful linear relationship between Apple returns and market returns during the analyzed period.

The estimated coefficient of determination of approximately $R^2 = 0.625$ indicates that the market factor alone explains a substantial fraction of the observed variability in Apple daily returns during the selected period. Nevertheless, a significant portion of the variability remains unexplained, motivating the consideration of multifactor extensions beyond the single-factor CAPM specification.

The fitted linear relationship between asset and market returns is illustrated in Figure 1.4.

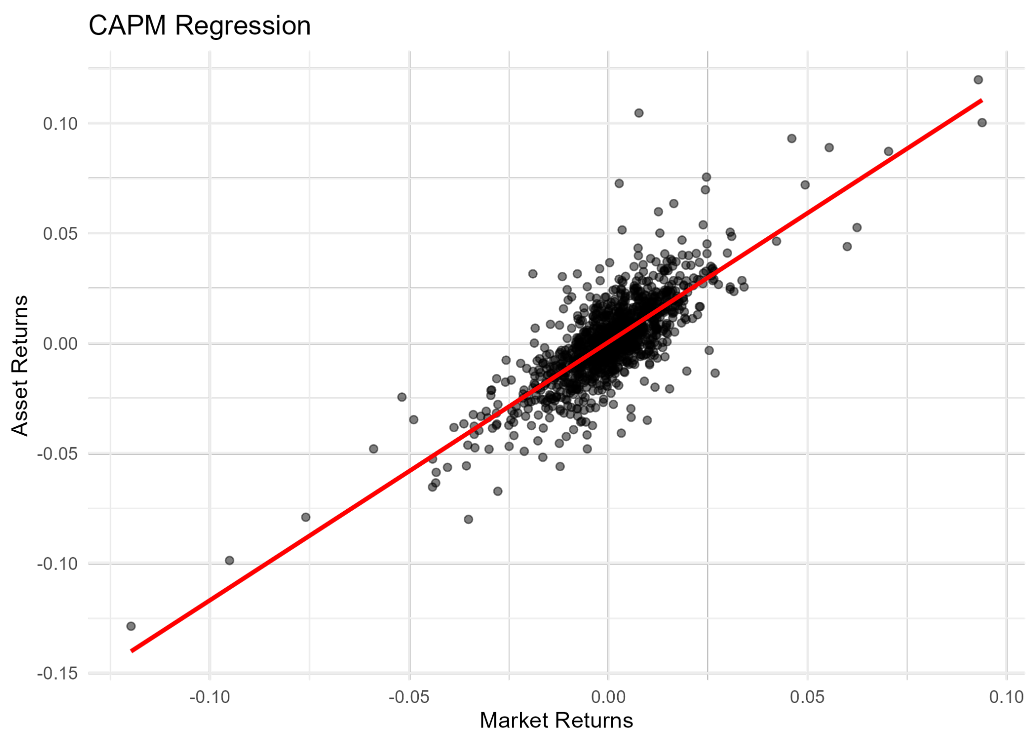

Figure 1.4: CAPM linear regression between Apple daily returns and S&P 500 daily returns. The red line represents the fitted linear relationship estimated through ordinary least squares regression.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

1.3.3 Fama–French Three-Factor Model Implementation

To extend the single-factor CAPM specification, the Fama–French three-factor model introduced in Equation 1.3 was estimated using ordinary least squares (OLS) regression. In addition to the market excess return factor, the model incorporates two additional systematic risk factors associated with firm size (`SMB`) and relative valuation (`HML`).

The implementation combines daily adjusted closing prices for Apple Inc. (`AAPL`) obtained from Yahoo Finance with the daily Fama–French factor dataset developed by Kenneth French. The final modeling dataset contains 1,256 daily observations covering the period from January 2020 to December 2024. The average daily excess return of Apple over the sample period was approximately 0.11%, with a daily standard deviation close to 2.0%.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

Figure 1.5 displays the behavior of the three explanatory factors over time. The series exhibit substantial short-term variability, particularly during periods of market stress such as the early stages of the COVID-19 pandemic in 2020. The `SMB` factor shows the highest volatility among the three factors during this period, reflecting the stronger sensitivity of small-cap firms to abrupt changes in market conditions.

The estimated regression coefficients are summarized below.

Table 1.3: Estimated Fama–French three-factor regression coefficients.

Term	Estimate	Std. Error	t-statistic	p-value
Alpha	0.000455	0.000320	1.421686	0.155366
Market Factor	1.162676	0.024001	48.442699	0.000000
Size Factor	-0.329641	0.042313	-7.790601	0.000000
Value Factor	-0.370591	0.027910	-13.278185	0.000000

The estimated market beta is approximately $\beta_{mkt} = 1.163$, indicating that Apple

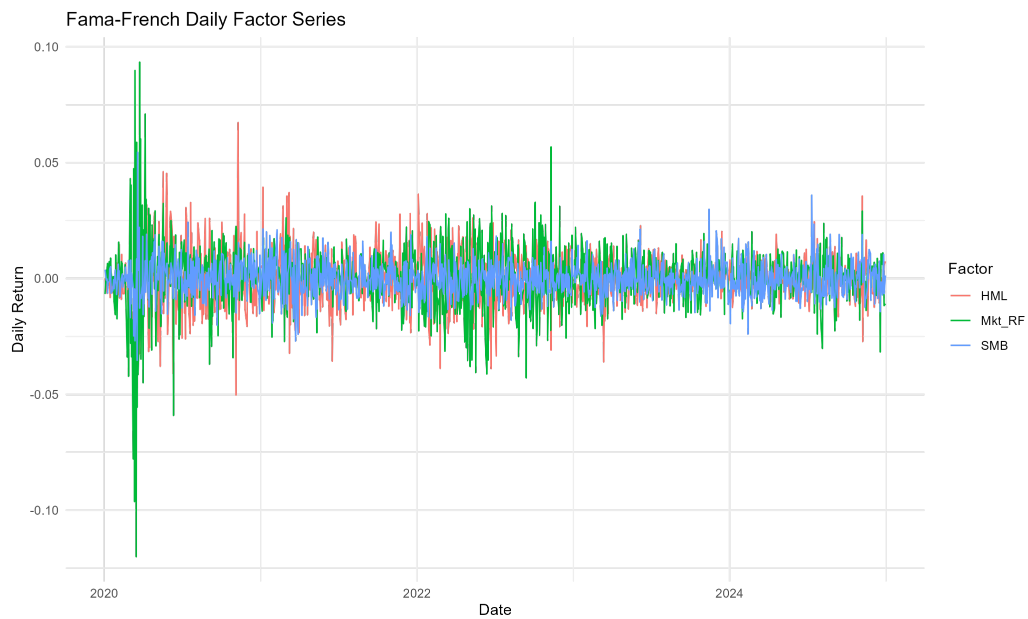

Figure 1.5: Fama–French daily factor series.

exhibits a stronger sensitivity to systematic market movements than the market portfolio itself. The coefficients associated with the size and value factors are both negative and statistically significant. The negative loading on `SMB` suggests that Apple behaves more similarly to large-cap firms than to small-cap firms, which is consistent with its market capitalization and dominant position within the technology sector. Likewise, the negative `HML` exposure indicates characteristics more closely aligned with growth-oriented firms rather than value stocks.

The estimated intercept term ($\alpha \approx 0.00046$) is positive but not statistically significant at conventional significance levels ($p \approx 0.155$), suggesting that, after accounting for the three systematic risk factors, the model does not provide strong evidence of abnormal excess returns during the analyzed period.

The overall regression fit improved relative to the single-factor CAPM specification. The coefficient of determination increased to approximately $R^2 = 0.678$, indicating that the three-factor model explains nearly 68% of the variability observed in Apple excess returns. The F-statistic associated with the regression is also highly significant, supporting the joint explanatory power of the selected factors.

Table 1.4: Diagnostic statistics associated with the Fama-French three-factor regression model.

Metric	Value
R-squared	0.677865
Adjusted R-squared	0.677093
Sigma	0.011345
F-statistic	878.191163
p-value	0.000000
Observations	1256

Figure 1.6 compares the observed excess returns against the fitted values generated by the regression model. The fitted series captures a substantial fraction of the short-term dynamics of Apple returns, although large residual fluctuations remain during periods of elevated volatility.

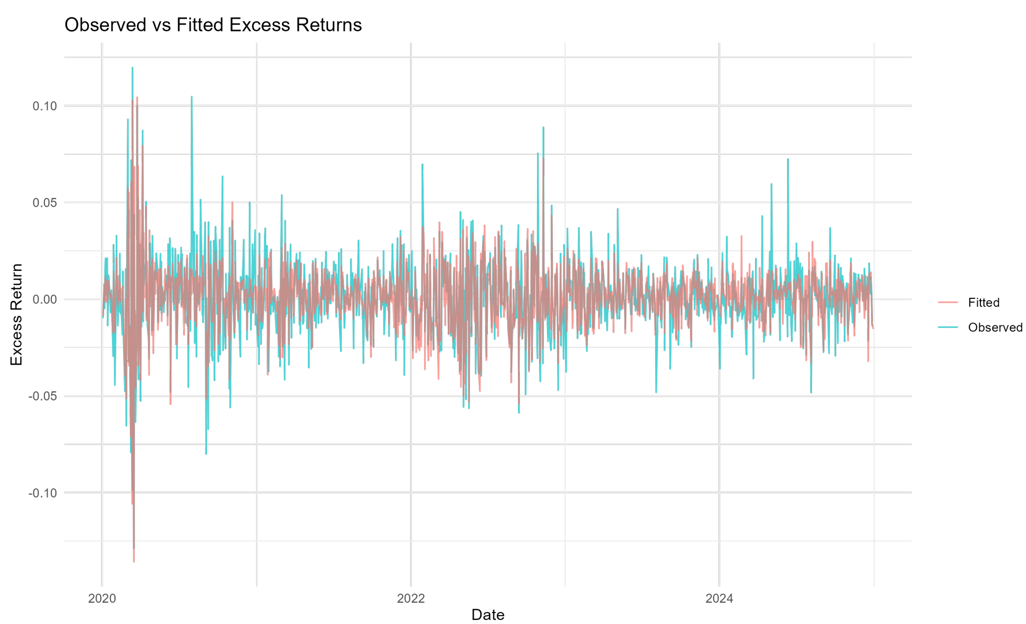

Figure 1.6: Observed and fitted excess returns under the Fama–French three-factor specification.

The residual series displayed in Figure 1.7 remains approximately centered around zero throughout the analyzed period, without obvious long-term structural drift. However, clusters of higher residual volatility are still visible during stressed market conditions, indicating that a portion of the variability remains unexplained even after incorporating the additional systematic factors.

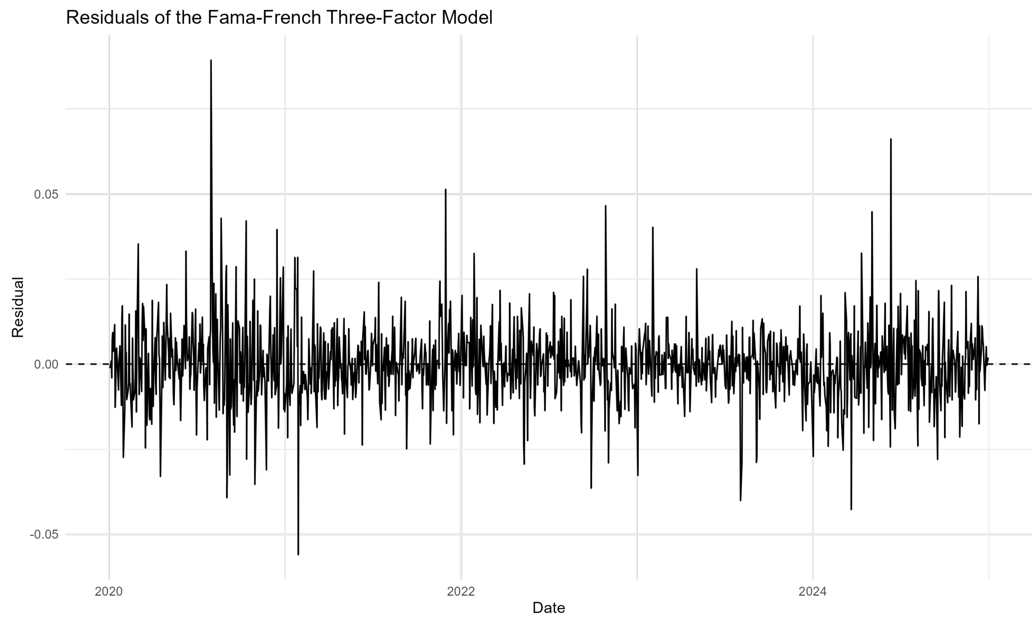

Figure 1.7: Residual series of the Fama–French three-factor model.

Finally, the residual distribution shown in Figure 1.8 exhibits a roughly symmetric shape centered near zero, although moderate tail behavior remains visible. This behavior is consistent with the presence of occasional large market movements that are not fully captured by the linear factor structure.

1.4 Results, Discussion, and Limitations

The empirical results obtained from both the CAPM and the Fama–French three-factor model provide a useful comparison between single-factor and multifactor approaches to asset pricing. Although both models capture an important fraction of the variability observed in Apple excess returns, the inclusion of additional systematic risk factors in the Fama–French specification produces a measurable improvement in explanatory performance.

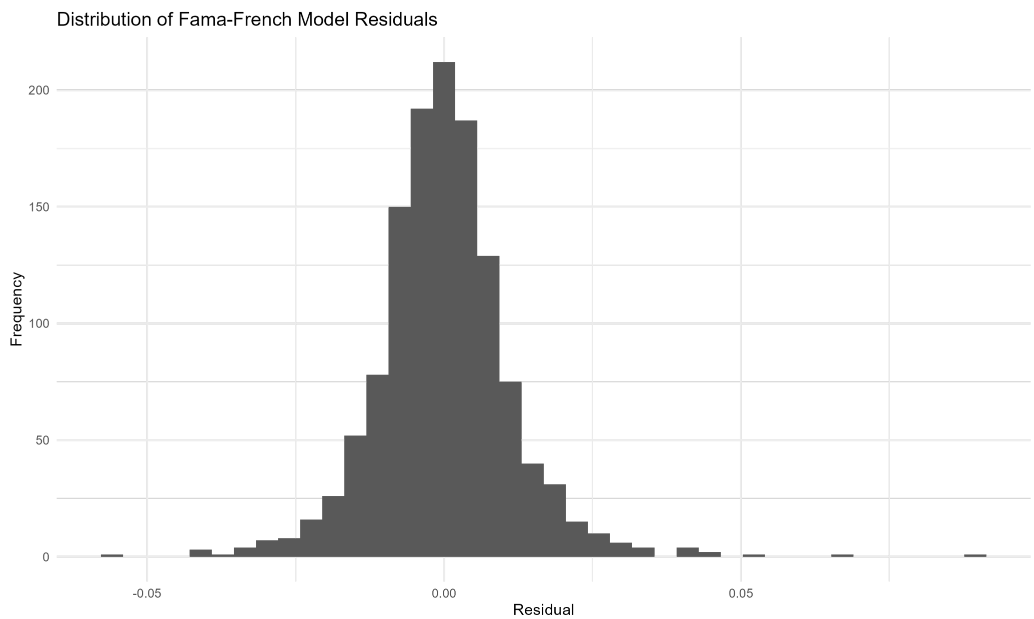

Figure 1.8: Distribution of residuals from the Fama–French three-factor model.

1.4.1 Comparative Analysis

The CAPM estimation produced a market beta close to $\beta \approx 1.17$, indicating that Apple exhibits a level of systematic risk slightly higher than that of the overall market portfolio. The market factor alone explained approximately 62.5% of the variability in Apple excess returns over the analyzed period, resulting in an estimated coefficient of determination of approximately $R^2 = 0.625$.

After incorporating the size (`SMB`) and value (`HML`) factors, the explanatory power of the regression increased to approximately $R^2 = 0.678$. Although the increase in explanatory performance is moderate, the multifactor specification captures an additional fraction of return variability that remains unexplained under the single-factor CAPM framework.

The estimated market beta remained relatively stable across both models, suggesting that market risk continues to represent the dominant systematic component driving Apple returns. However, the additional factor loadings provide further insight into the firm's structural characteristics. The negative exposure to the size factor indicates that Apple behaves more similarly to large-cap firms than to smaller companies, while the negative loading associated with the value factor reflects characteristics commonly associated with growth-oriented firms.

From a statistical perspective, both models produced highly significant overall regressions according to their respective F-statistics. Nevertheless, the Fama–French specification generated lower residual variability and a modest improvement in fitted return dynamics, particularly during periods of elevated market volatility.

The comparison between observed and fitted excess returns also suggests that the multifactor specification is better able to reproduce short-term return fluctuations than the CAPM alone. However, significant residual movements remain visible in both models, indicating that a substantial fraction of the variability in daily equity returns cannot be fully captured through linear factor structures alone.

Overall, the empirical results support the interpretation that the Fama–French three-factor model provides a richer and more flexible representation of systematic risk exposures relative to the standard CAPM specification. At the same time, the relatively modest improvement in explanatory power also highlights the dominant role played by market-wide risk in large-cap technology firms such as Apple.

1.4.2 Model Limitations

Despite their widespread use in financial economics, both the CAPM and the Fama–French three-factor model rely on simplifying assumptions that limit their ability to fully explain observed market behavior.

First, both models assume linear relationships between excess returns and systematic risk factors. In practice, financial markets frequently exhibit nonlinear dynamics, volatility clustering, structural breaks, and regime-dependent behavior that cannot be fully represented within a static linear regression framework.

Second, the analysis relies on historical daily observations and therefore assumes that the estimated factor sensitivities remain relatively stable throughout the selected time horizon. However, the relationship between individual assets and systematic risk factors may evolve over time as market conditions, macroeconomic environments, or firm-specific characteristics change.

An additional limitation arises from the selection of explanatory factors themselves. Although the Fama–French specification extends the CAPM by incorporating size and value effects, many additional sources of systematic risk have been proposed in the literature, including profitability, investment, momentum, liquidity, and volatility-related factors. Consequently, the three-factor model should not be interpreted as a complete description of equity return dynamics.

The analysis also focuses exclusively on a single large-cap technology company. While Apple represents an interesting case study due to its market relevance and liquidity, the estimated coefficients and model diagnostics cannot be directly generalized to other sectors, asset classes, or market environments without additional empirical validation.

Finally, both models are estimated under the assumption that historical relationships provide meaningful information regarding future risk-return behavior. As in most empirical financial applications, this assumption may fail under changing market conditions, particularly during periods of financial stress or structural economic transitions.

Despite these limitations, the CAPM and Fama–French frameworks continue to provide valuable benchmark models for understanding systematic risk exposures, comparing relative asset sensitivities, and constructing interpretable factor-based representations of financial returns.

1.5 Technical Appendix (Optional)

This appendix provides a compact technical background for the regression models used in this project. The main body of the chapter focuses on the financial interpretation of the CAPM and Fama-French models, while this appendix summarizes the statistical structure underlying their empirical estimation.

The CAPM specification can be interpreted as a simple linear regression model, where the excess return of an asset is explained by the excess return of the market portfolio. In contrast, the Fama-French three-factor model extends this idea to a multiple linear regression setting, where asset excess returns are explained by a set of risk factors rather than by the market factor alone.

The purpose of this appendix is not to provide a complete treatment of linear regression or statistical learning. Instead, it collects the main concepts needed to understand how the models were estimated, how the coefficients should be interpreted, and how goodness-of-fit and statistical significance measures relate to the empirical results discussed in the project. The presentation follows standard treatments of linear regression and statistical learning [9] [10].

1.5.1 Simple Linear Regression and the CAPM Model

As discussed previously in Equation 1.2, the CAPM establishes a linear relationship between the expected excess return of an asset and the expected excess return of the market portfolio. From a statistical perspective, this relationship can be interpreted as a simple linear regression model of the form

$$y_i = \theta_0 + \theta_1 x_i + \varepsilon_i \tag{1.4}$$

where θ_0 represents the intercept, θ_1 the slope coefficient, and ε_i an unobservable random error associated with observation i.

Under this interpretation, the empirical CAPM specification can be written as

$$r_i - r_f = \alpha + \beta(r_m - r_f) + \varepsilon \tag{1.5}$$

where the excess return of the asset plays the role of the response variable and the excess return of the market portfolio acts as the predictor variable. In this setting, α corresponds to the intercept term, while β measures the sensitivity of the asset return to market movements.

From the perspective of the CAPM, the intercept term α is expected to be equal to zero, implying that expected excess returns are fully explained by exposure to market risk through the coefficient β. Consequently, testing whether α differs significantly from zero provides a practical way to evaluate the empirical consistency of the model.

1.5.1.1 Ordinary Least Squares Estimation

The parameters of the regression model are estimated by minimizing the discrepancy between the observed data and the fitted regression line. In ordinary least squares (OLS), this discrepancy is quantified through the residual sum of squares (RSS), defined as

$$\text{RSS} = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{N} (y_i - \theta_0 - \theta_1 x_i)^2 \tag{1.6}$$

where each residual represents the difference between an observed value and the value predicted by the model.

Figure 1.9 illustrates the geometric interpretation of the residuals in simple linear regression. The ordinary least squares estimator selects the parameters that minimize the squared residual distances between the observations and the fitted regression line.

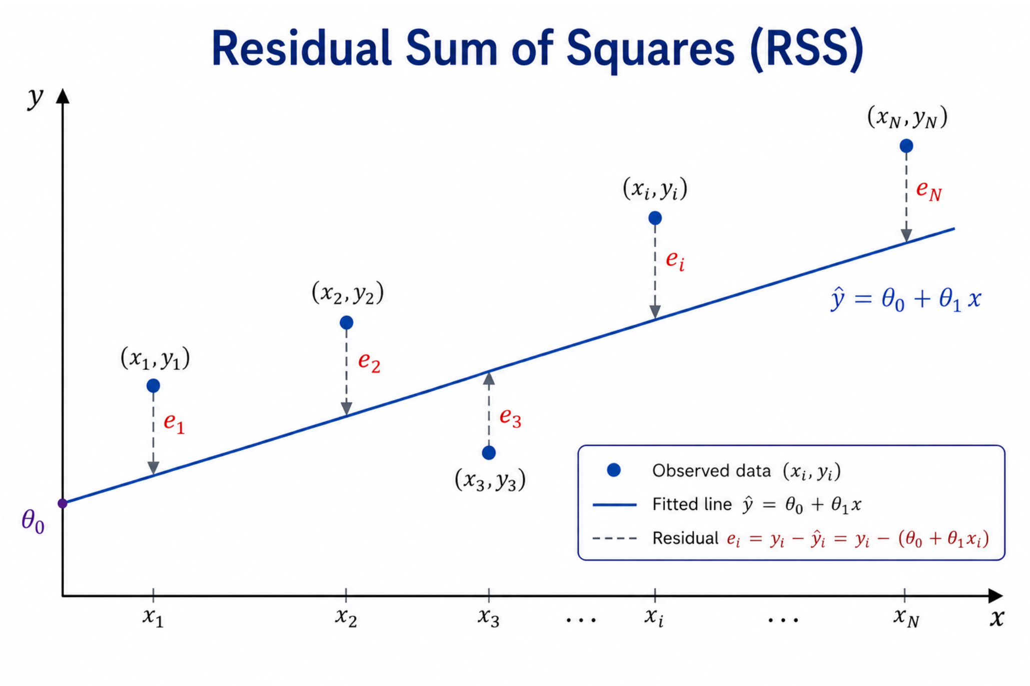

Figure 1.9: Residual Sum of Squares (RSS)

The ordinary least squares estimators are obtained by minimizing Equation 1.6 with respect to the model parameters. Since the RSS is a convex quadratic function of θ_0 and θ_1, the solution can be obtained by setting the gradient equal to zero. The convexity of the RSS follows from the positive definiteness of the associated Hessian matrix, ensuring that the stationary point corresponds to a unique global minimum.

The resulting estimators are given by

$$\hat{\theta}_1 = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})^2} = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)} \tag{1.7}$$

and

$$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \tag{1.8}$$

Equation 1.7 shows that the slope estimator can be interpreted as the ratio between the covariance of x and y and the variance of x. Consequently, the estimated slope increases when both variables tend to move together and decreases when their linear relationship weakens. In the context of empirical asset pricing, this interpretation leads naturally to the CAPM market beta, which can be viewed as the ordinary least squares estimator obtained by regressing the excess return of an asset against the excess return of the market portfolio. Under this interpretation, the CAPM beta measures the linear sensitivity of the asset return to market movements in the least-squares sense.

1.5.1.2 Residual Analysis and Model Assumptions

Once the regression coefficients have been estimated, the residuals provide information about the portion of the variability that is not explained by the model. For each observation, the residual is defined as

$$e_i = y_i - \hat{y}_i \tag{1.9}$$

where $\hat{y}_i$ denotes the predicted value obtained from the fitted regression model.

A distinction should be made between residuals and random errors. Residuals correspond to observable deviations between the fitted model and the data, whereas the random errors ε_i introduced in Equation 1.4 represent the unobservable stochastic component assumed by the statistical model.

Classical linear regression assumes that the random errors are independent, centered around zero, and have constant variance. Under these assumptions,

$$\varepsilon_i \sim N(0, \sigma^2) \tag{1.10}$$

where σ^2 denotes the variance of the error term.

The assumption of constant variance is commonly referred to as homoscedasticity and can be expressed as

$$\mathrm{Var}(\varepsilon_i) = \sigma^2 \tag{1.11}$$

for all observations. When the variance of the errors changes across observations, the model is said to exhibit heteroscedasticity, which may affect the reliability of confidence intervals, hypothesis tests, and statistical inference.

Although these assumptions are rarely satisfied exactly in real financial data, they provide the statistical framework underlying ordinary least squares estimation and the interpretation of the inferential quantities discussed in the following sections.

1.5.1.3 Confidence Intervals and Standard Errors

The regression coefficients estimated through ordinary least squares are subject to sampling variability. Consequently, point estimates alone do not provide information about the uncertainty associated with the estimated parameters. This uncertainty is commonly quantified through standard errors and confidence intervals.

Under the assumptions introduced previously, the variance of the random error term can be estimated from the residuals through the residual standard error (RSE), defined as

$$\mathrm{RSE} = \sqrt{\frac{\mathrm{RSS}}{N-2}} \tag{1.12}$$

where N denotes the number of observations and the subtraction of two degrees of freedom reflects the estimation of the intercept and slope coefficients.

The standard errors associated with the regression coefficients are given by

$$\mathrm{SE}(\hat{\theta}_1) = \sqrt{\frac{\sigma^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}} \tag{1.13}$$

and

$$\mathrm{SE}(\hat{\theta}_0) = \sqrt{\sigma^2 \left[\frac{1}{N} + \frac{\bar{x}^2}{\sum_{i=1}^{N}(x_i - \bar{x})^2}\right]} \tag{1.14}$$

where σ^2 is typically replaced in practice by the estimator introduced in Equation 1.12.

These expressions show that the uncertainty associated with the estimated coefficients depends on both the residual variability and the dispersion of the predictor variable. In particular, the standard error of the slope coefficient decreases as the variability of x increases, leading to more stable estimates of the linear relationship.

Confidence intervals provide a range of plausible values for the regression coefficients. Under approximate normality assumptions, a 95% confidence interval for a parameter θ can be written as

$$\hat{\theta} \pm 2 \cdot \mathrm{SE}(\hat{\theta}) \tag{1.15}$$

indicating the range within which the true parameter value is expected to lie with approximately 95% confidence.

1.5.1.4 Hypothesis Testing and Statistical Significance

Beyond estimating the regression coefficients, it is often important to evaluate whether the observed linear relationship is statistically significant. In simple linear regression, this is commonly assessed through hypothesis testing on the slope coefficient.

The null and alternative hypotheses are given by

$$\begin{cases} H_0 : \theta_1 = 0 \\ H_1 : \theta_1 \neq 0 \end{cases} \tag{1.16}$$

Under the null hypothesis, the predictor variable does not contribute to explaining the variability of the response variable, implying the absence of a linear relationship between x and y.

The statistical significance of the estimated slope coefficient can be evaluated through the t-statistic,

$$t = \frac{\hat{\theta}_1}{\mathrm{SE}(\hat{\theta}_1)} \tag{1.17}$$

which measures the magnitude of the estimated coefficient relative to its standard error.

The corresponding p-value quantifies the probability of observing a statistic at least as extreme as the one obtained under the null hypothesis. Small p-values provide evidence against H_0, suggesting that the predictor variable contributes significantly to the regression model.

In the context of the CAPM, hypothesis testing is particularly relevant for evaluating whether the estimated market exposure coefficient differs significantly from zero and whether the intercept term is statistically consistent with the theoretical prediction of $\alpha = 0$.

1.5.1.5 Analysis of Variance (ANOVA)

The variability of the response variable can be decomposed into a component explained by the regression model and a residual component associated with unexplained variability. In simple linear regression, this decomposition is expressed as

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i) \tag{1.18}$$

which separates the total deviation of each observation into an explained component and a residual component.

Figure 1.10 illustrates the geometric interpretation of this decomposition in the context of simple linear regression.

Summing over all observations leads to the classical ANOVA decomposition

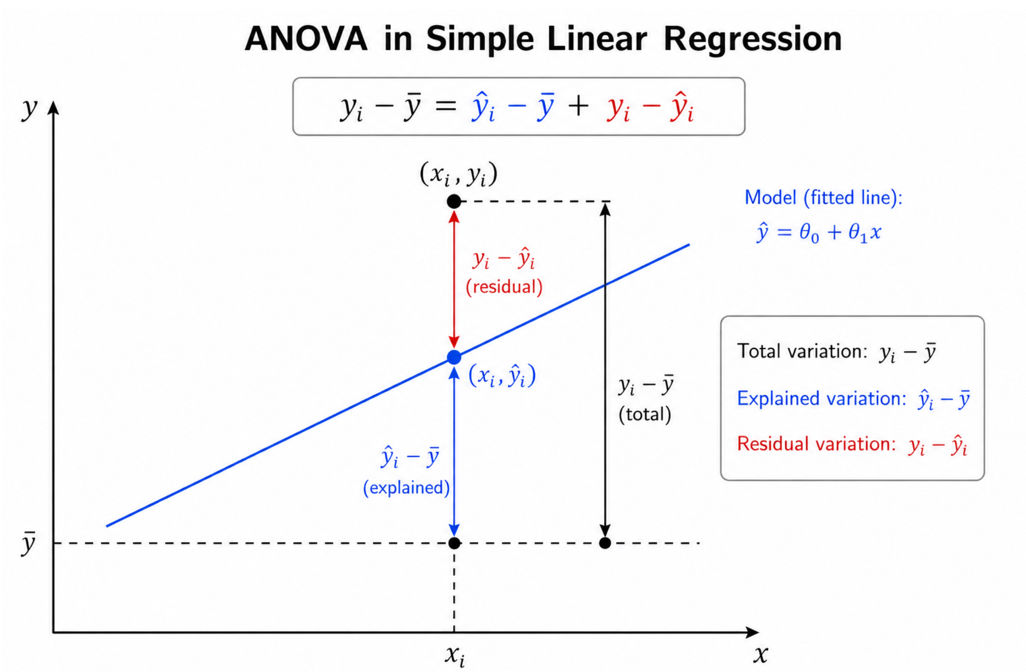

Figure 1.10: ANOVA decomposition in simple linear regression

$$\text{TSS} = \text{MSS} + \text{RSS} \tag{1.19}$$

where

$$\text{TSS} = \sum_{i=1}^{N} (y_i - \bar{y})^2 \tag{1.20}$$

represents the total variability of the response variable,

$$\text{MSS} = \sum_{i=1}^{N} (\hat{y}_i - \bar{y})^2 \tag{1.21}$$

corresponds to the variability explained by the regression model, and

$$\text{RSS} = \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{1.22}$$

measures the unexplained residual variability.

Dividing each sum of squares by its associated degrees of freedom leads to the corresponding mean squares. In simple linear regression, the mean model square and the mean residual square are given by

$$\text{MMS} = \text{MSS} \tag{1.23}$$

and

$$\text{MSR} = \frac{\text{RSS}}{N-2} \tag{1.24}$$

respectively.

These quantities allow the construction of the F-statistic,

$$F = \frac{\text{MMS}}{\text{MSR}} \tag{1.25}$$

which compares the variability explained by the regression model with the unexplained residual variability. Large values of the F-statistic suggest that the regression model explains a substantial fraction of the observed variability relative to the residual noise.

1.5.1.6 Coefficient of Determination (R^2)

Another commonly used measure in linear regression is the coefficient of determination, denoted by R^2. This statistic quantifies the proportion of the total variability in the response variable that is explained by the regression model.

Using the ANOVA decomposition introduced in Equation 1.19,

$$\text{TSS} = \text{MSS} + \text{RSS}$$

the coefficient of determination can be written as

$$R^2 = \frac{\text{MSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \tag{1.26}$$

Larger values of R^2 indicate that the model explains a greater fraction of the observed variability in the data. In particular, $R^2 = 1$ corresponds to a perfect fit, whereas $R^2 = 0$ indicates that the regression model does not explain the variability of the response variable better than the sample mean alone.

In the special case of simple linear regression, the coefficient of determination is directly related to the Pearson correlation coefficient through

$$R^2 = r^2 \tag{1.27}$$

where r denotes the sample correlation between the predictor variable x and the response variable y.

Although R^2 provides a useful measure of goodness of fit, a large value does not necessarily imply that the model is statistically valid or economically meaningful. For this reason, R^2 is typically interpreted together with residual diagnostics, hypothesis tests, and domain-specific considerations.

1.5.2 Multiple Linear Regression and the Fama-French Model

While the CAPM can be formulated as a simple linear regression model with a single explanatory factor, the Fama–French three-factor model extends this framework by incorporating additional sources of systematic risk. From a statistical perspective, this leads naturally to a multiple linear regression model in which the response variable depends simultaneously on several predictors.

For a model with M explanatory variables, the linear regression model can be written as

$$y_i = \theta_0 + \theta_1 x_{i1} + \theta_2 x_{i2} + \cdots + \theta_M x_{iM} + \varepsilon_i \tag{1.28}$$

where θ_0 represents the intercept, $\theta_1, \dots, \theta_M$ denote the regression coefficients associated with the predictor variables, and ε_i corresponds to the random error associated with observation i.

Using matrix notation, the model can be expressed compactly as

$$\mathbf{y} = \mathbf{X}\theta + \varepsilon \tag{1.29}$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1M} \\ 1 & x_{21} & \cdots & x_{2M} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{NM} \end{bmatrix}$$

and

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_M \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_N \end{bmatrix}$$

In this representation, the rows of $\mathbf{X}$ correspond to observations, while the columns represent the explanatory variables included in the model. The matrix formulation provides a compact and computationally efficient framework for parameter estimation and statistical inference.

Under this interpretation, the Fama–French three-factor model can be viewed as a particular case of multiple linear regression in which the excess return of an asset is explained simultaneously by the market factor, the SMB factor, and the HML factor.

1.5.2.1 Ordinary Least Squares Estimation

As in the simple linear regression setting, the parameters of the multiple linear regression model are estimated by minimizing the residual sum of squares. Using the matrix formulation introduced in Equation 1.29, the residual vector can be written as

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\theta \tag{1.30}$$

and the corresponding residual sum of squares is given by

$$\mathrm{RSS} = \mathbf{e}^T\mathbf{e} = (\mathbf{y} - \mathbf{X}\theta)^T(\mathbf{y} - \mathbf{X}\theta) \tag{1.31}$$

The ordinary least squares estimator is obtained by minimizing Equation 1.31 with respect to the parameter vector θ. As in the simple linear regression case, the optimization problem is convex and admits a unique global minimum provided that the columns of the design matrix $\mathbf{X}$ are linearly independent.

The resulting estimator is given by

$$\hat{\theta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{1.32}$$

which generalizes the ordinary least squares estimator introduced previously for simple linear regression.

The matrix formulation highlights an important aspect of multiple linear regression: each coefficient measures the linear contribution of its associated predictor while accounting for the remaining variables included in the model. In the context of the Fama–French model, this means that the estimated coefficients associated with the market, SMB, and HML factors represent their partial contributions to the excess return of the asset after controlling for the other factors included in the regression.

1.5.2.2 Statistical Interpretation of the Coefficients

In multiple linear regression, each regression coefficient measures the expected variation in the response variable associated with a one-unit change in the corresponding predictor, while holding the remaining predictors fixed. This interpretation differs from simple linear regression, where the effect of the predictor is evaluated without conditioning on additional explanatory variables.

Under the ordinary least squares framework, the uncertainty associated with the estimated parameter vector can be characterized through its covariance matrix, given by

$$\mathrm{Cov}(\hat{\theta}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} \tag{1.33}$$

where σ^2 denotes the variance of the random error term.

As in the simple linear regression setting, the variance of the residuals can be estimated through the residual standard error,

$$\mathrm{RSE} = \sqrt{\frac{\mathrm{RSS}}{N - M - 1}} \tag{1.34}$$

where N represents the number of observations and M the number of explanatory variables included in the model.

The diagonal elements of Equation 1.33 determine the standard errors associated with each regression coefficient and provide a measure of the uncertainty of the parameter estimates. Larger standard errors indicate greater uncertainty in the estimated contribution of the corresponding predictor.

In the context of the Fama–French three-factor model, the estimated coefficients associated with the market, SMB, and HML factors measure the sensitivity of the asset return to each systematic risk factor after controlling for the remaining factors included in the regression. Consequently, the interpretation of each coefficient should always be understood conditionally on the presence of the other explanatory variables.

The inclusion of multiple predictors may also introduce statistical challenges such as multicollinearity, which occurs when explanatory variables exhibit strong linear dependence. In such cases, coefficient estimates may become unstable and difficult to interpret, even when the overall predictive performance of the model remains satisfactory.

1.5.2.3 Global Hypothesis Testing and Model Significance

In multiple linear regression, it is often important to evaluate whether the explanatory variables collectively contribute to the model. This can be assessed through a global hypothesis test of the form

$$\begin{cases} H_0 : \theta_1 = \theta_2 = \cdots = \theta_M = 0 \\ H_1 : \text{at least one } \theta_j \neq 0 \end{cases} \tag{1.35}$$

Under the null hypothesis, none of the explanatory variables contribute to explaining the variability of the response variable, implying that the model does not perform better than using the sample mean alone.

The global significance of the regression model can be evaluated through the F-statistic,

$$F = \frac{(\text{TSS} - \text{RSS})/M}{\text{RSS}/(N - M - 1)} \tag{1.36}$$

which compares the variability explained by the model with the unexplained residual variability after adjusting for the corresponding degrees of freedom.

Large values of the F-statistic suggest that the explanatory variables collectively explain a substantial fraction of the observed variability in the response variable. The corresponding p-value provides a quantitative measure of the evidence against the null hypothesis.

In the context of the Fama–French three-factor model, the global F-test evaluates whether the market, SMB, and HML factors jointly contribute to explaining the excess return of the asset. Consequently, the test provides a global assessment of the explanatory power of the multifactor specification relative to a model without risk factors.

1.5.2.4 Limitations of Multiple Linear Regression

Although multiple linear regression provides a flexible and interpretable framework for modeling relationships between variables, its performance depends on several assumptions and modeling choices. In practice, violations of these assumptions may affect the reliability of parameter estimates, confidence intervals, and hypothesis tests.

One important limitation is multicollinearity, which arises when explanatory variables exhibit strong linear dependence. In such cases, the regression coefficients may become unstable and highly sensitive to small changes in the data, making their interpretation more difficult.

Multiple linear regression models may also be affected by omitted variable bias when relevant explanatory factors are excluded from the specification. In empirical asset pricing, this issue motivated the development of multifactor models such as the Fama–French framework, which extends the CAPM by incorporating additional sources of systematic risk beyond the market factor alone.

Another practical limitation is that linear regression assumes a linear relationship between predictors and the response variable. While this assumption often provides a useful approximation, real financial data may exhibit nonlinear effects, structural changes, heteroscedasticity, or temporal dependence that are not fully captured by the model.

Despite these limitations, linear regression models remain fundamental tools in empirical finance due to their interpretability, computational efficiency, and strong connection with statistical inference and risk factor modeling.

Chapter 2

Unsupervised Learning for Asset Allocation

This project explores the use of unsupervised learning techniques for portfolio construction and risk diversification. The analysis combines Principal Component Analysis (PCA), hierarchical clustering, and Hierarchical Risk Parity (HRP) to study how latent market structure and correlation patterns can be used to build more stable asset allocation strategies.

Rather than focusing on return forecasting, the project examines how statistical learning methods can help identify common sources of risk, reduce redundancy across correlated assets, and improve portfolio robustness under realistic market conditions. The underlying idea is that portfolios are often considered diversified based on asset labels, sectors, or asset classes, even when a small number of common risk factors continue to drive most of their behaviour. By studying how assets actually move together, the project aims to construct allocations that better reflect the underlying structure of market risk.

2.1 Problem Definition

Constructing a diversified investment portfolio is one of the central problems in quantitative finance. In practice, investors must allocate capital across a large universe of financial instruments while balancing expected returns, risk exposure, diversification, and portfolio stability. Although the classical mean–

variance framework introduced by Harry Markowitz remains foundational in modern portfolio theory, its practical implementation presents important limitations when applied to real financial markets.

Traditional portfolio optimization methods rely heavily on the estimation and inversion of covariance matrices. In large and highly correlated markets, these matrices often become numerically unstable and highly sensitive to small estimation errors. As a result, optimized portfolios may exhibit unintuitive allocations, excessive concentration, poor robustness, and unstable out-of-sample performance. This issue becomes more severe as the number of assets increases and correlation structures evolve through time.

At the same time, financial markets naturally exhibit latent and hierarchical structures that are not fully captured by traditional optimization approaches. Assets rarely behave as isolated entities. Instead, they tend to move collectively according to sector dynamics, macroeconomic conditions, liquidity regimes, or broader market factors. Under stressed market conditions, assets that appear diversified on paper may suddenly behave in remarkably similar ways, reducing diversification precisely when it becomes most important.

The motivation behind this project is to explore whether unsupervised learning techniques can provide a more robust and structurally stable framework for asset allocation. Rather than treating portfolio construction purely as a constrained optimization problem, the project approaches it as a problem of identifying and organizing the underlying geometry of market risk.

To address this, the analysis combines two complementary perspectives. First, Principal Component Analysis (PCA) is used to study the dominant sources of variation in the market and to represent the covariance structure in terms of orthogonal risk components. This provides a compact representation of the market structure while reducing redundancy across highly correlated assets. Second, hierarchical clustering techniques are used to identify groups of statistically similar assets and to construct portfolios using Hierarchical Risk Parity (HRP), a methodology designed to improve allocation stability without requiring direct inversion of the covariance matrix.

The objective is not simply to maximize historical returns, but to investigate whether combining spectral methods and hierarchical structures can produce portfolios that are more diversified, interpretable, and robust under changing market conditions. The project therefore sits at the intersection of quantitative finance, machine learning, and statistical risk modeling, with a strong emphasis on practical portfolio construction rather than purely theoretical optimization.

2.2 Modeling Approach

The modeling approach adopted in this project treats portfolio construction as a problem of understanding the structure of market risk. The analysis focuses on how assets move together, how risk concentrates across the market, and how correlation patterns affect diversification under changing market conditions.

Financial markets are rarely composed of independent assets. Securities that belong to different sectors or asset classes often remain exposed to the same underlying drivers, particularly during periods of stress when correlations tend to increase. This means that portfolios that appear diversified on paper may still contain significant hidden concentrations of risk.

The analysis begins with Principal Component Analysis (PCA), following the general perspective of statistical risk decomposition discussed in [11] and the idea of principal portfolios introduced in [12]. PCA is used to study the covariance structure of the market and identify the dominant directions of variation across asset returns. In practice, a relatively small number of components often explains a large proportion of total market variance, suggesting that many assets continue to react to common underlying factors even when they appear superficially diversified.

The covariance matrix is decomposed according to

$$\Sigma = Q\Lambda Q^\top \tag{2.1}$$

where Q contains the eigenvectors and Λ the associated eigenvalues. The eigenvectors define orthogonal directions in the transformed risk space, while the eigenvalues measure the variance explained by each component. This decomposition provides a compact representation of the market structure and helps distinguish between broad market behaviour and more localized sources of variation.

The PCA transformation can also be interpreted as a rotation of the original return space into an orthogonal basis of uncorrelated components,

$$\tilde{\mathbf{R}} = \Gamma^{\top}\mathbf{R} \tag{2.2}$$

where $\mathbf{R}$ denotes the original return vector and $\tilde{\mathbf{R}}$ its representation in the transformed PCA space. Portfolio returns preserve their underlying structure under this transformation while becoming easier to interpret in terms of independent sources of market variation.

PCA is used as a diagnostic tool for studying market structure before constructing the final allocation. The portfolio construction stage is based on Hierarchical Risk Parity (HRP), introduced in [13] and further developed in [14]. Unlike traditional mean–variance optimization frameworks, HRP does not require direct inversion of the covariance matrix, avoiding part of the numerical instability associated with highly correlated asset universes and estimation error.

The clustering stage identifies groups of statistically similar assets using a correlation-based distance measure,

$$d_{ij} = \sqrt{\frac{1}{2}(1 - \rho_{ij})} \tag{2.3}$$

where ρ_{ij} denotes the correlation between assets i and j. Assets with similar behaviour are grouped together into a hierarchical structure that reflects the organization of the market more naturally than a flat covariance matrix alone.

Portfolio weights are then allocated recursively across the hierarchy, reducing concentration in highly correlated regions of the market. In practice, this tends to produce more stable allocations than traditional mean–variance optimization procedures, where small changes in covariance estimates can generate disproportionately large changes in portfolio weights.

An equally weighted $1/N$ portfolio is used as a benchmark throughout the analysis. Although simple, this benchmark provides an important reference point when evaluating whether additional model complexity leads to meaningful improvements in diversification and portfolio robustness out-of-sample.

The resulting framework combines spectral analysis, clustering techniques, and hierarchical risk allocation to study diversification from a structural perspective, with emphasis on robustness and stability under changing market conditions.

2.3 Methodology and Implementation

The empirical analysis was conducted using a diversified universe of U.S. equities representing multiple sectors of the economy, including technology, financials, energy, healthcare, consumer staples, communication services, and retail. The selected assets were chosen to provide exposure to different market dynamics while still allowing meaningful correlation structures and latent factor relationships to emerge throughout the analysis.

The asset universe used in the study is summarized in Table 2.1.

Table 2.1: Selected asset universe used throughout the portfolio construction and clustering experiments.

Ticker	Sector
AAPL	Technology
MSFT	Technology
GOOGL	Communication Services
AMZN	Consumer Discretionary
META	Communication Services
JPM	Financials
BAC	Financials
XOM	Energy
CVX	Energy
JNJ	Healthcare
PFE	Healthcare
PG	Consumer Staples
KO	Consumer Staples
WMT	Retail
HD	Consumer Discretionary

Daily adjusted closing prices were collected and transformed into continuously compounded daily returns. The analysis focused primarily on the covariance and correlation structure of returns rather than on absolute price levels. Figure 2.1 presents the evolution of adjusted prices across the selected assets during the evaluation period.

Although the selected securities belong to different sectors, the return series ex-

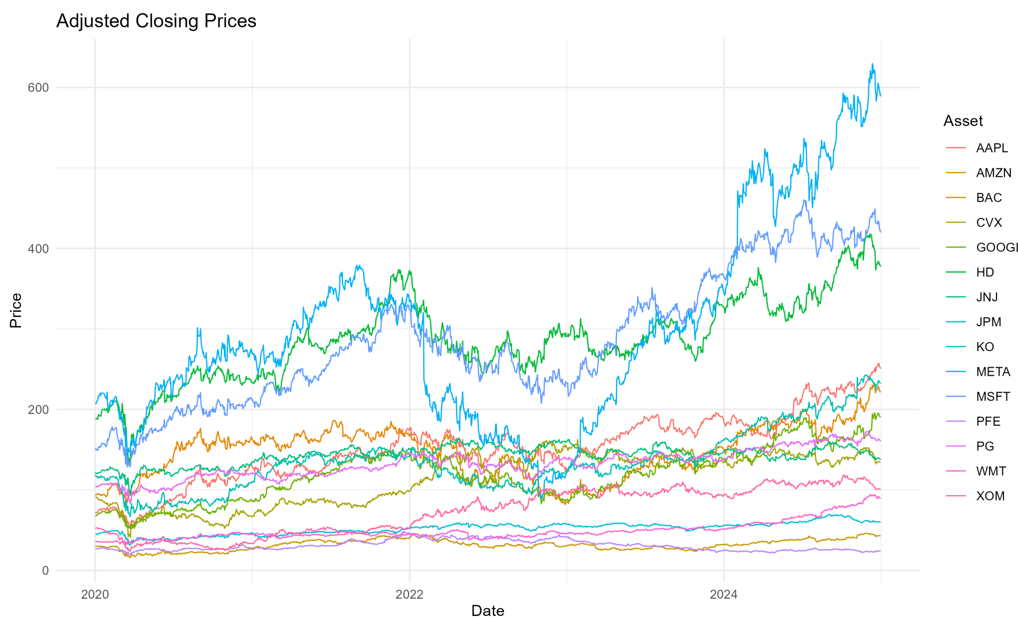

Figure 2.1: Adjusted closing prices of the selected assets over the analysis period, illustrating heterogeneous price dynamics and sector-specific trends across the asset universe.

hibit periods of synchronized behaviour, particularly during episodes of elevated market stress. This becomes more evident when examining the daily return dynamics shown in Figure 2.2.

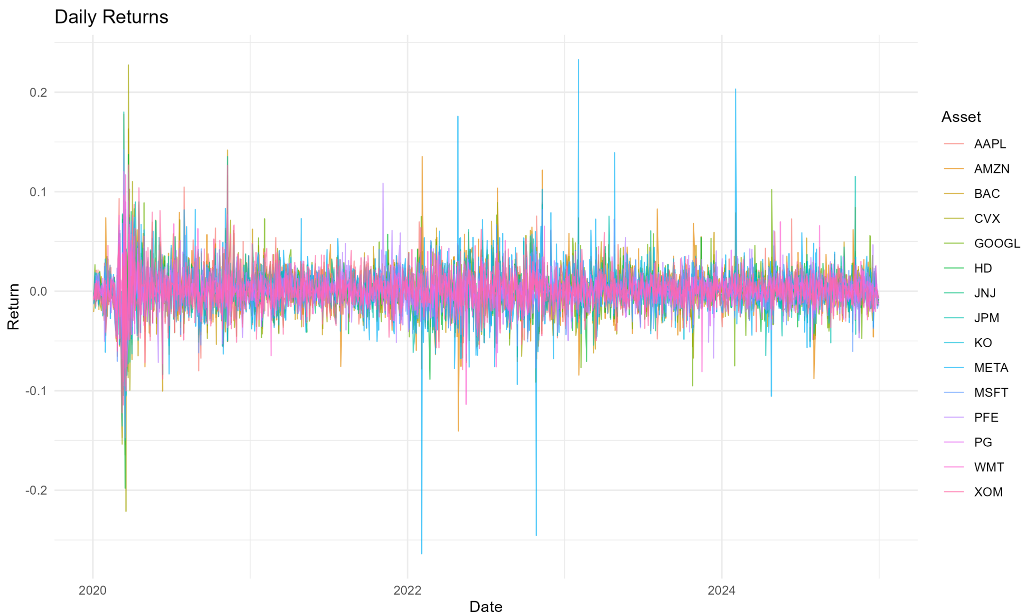

Figure 2.2: Daily returns of the selected assets over the analysis period, highlighting volatility clustering and heterogeneous risk dynamics across the asset universe.

The return series display the volatility clustering commonly observed in financial markets, together with periods of sharply increasing cross-asset dependence. The volatility spikes and correlation increases visible in the data are precisely the conditions under which naïve diversification can become less effective, providing the empirical motivation for the PCA and HRP framework adopted in this study.

The covariance matrix shown in Figure 2.3 was estimated directly from the daily return series and used throughout the PCA and HRP stages of the analysis.

The corresponding correlation matrix is presented in Figure 4.5. Several economically meaningful relationships already become visible at this stage, particularly among technology, financial, energy, healthcare, and consumer staples stocks.

Principal Component Analysis (PCA) was then applied to study the latent struc-

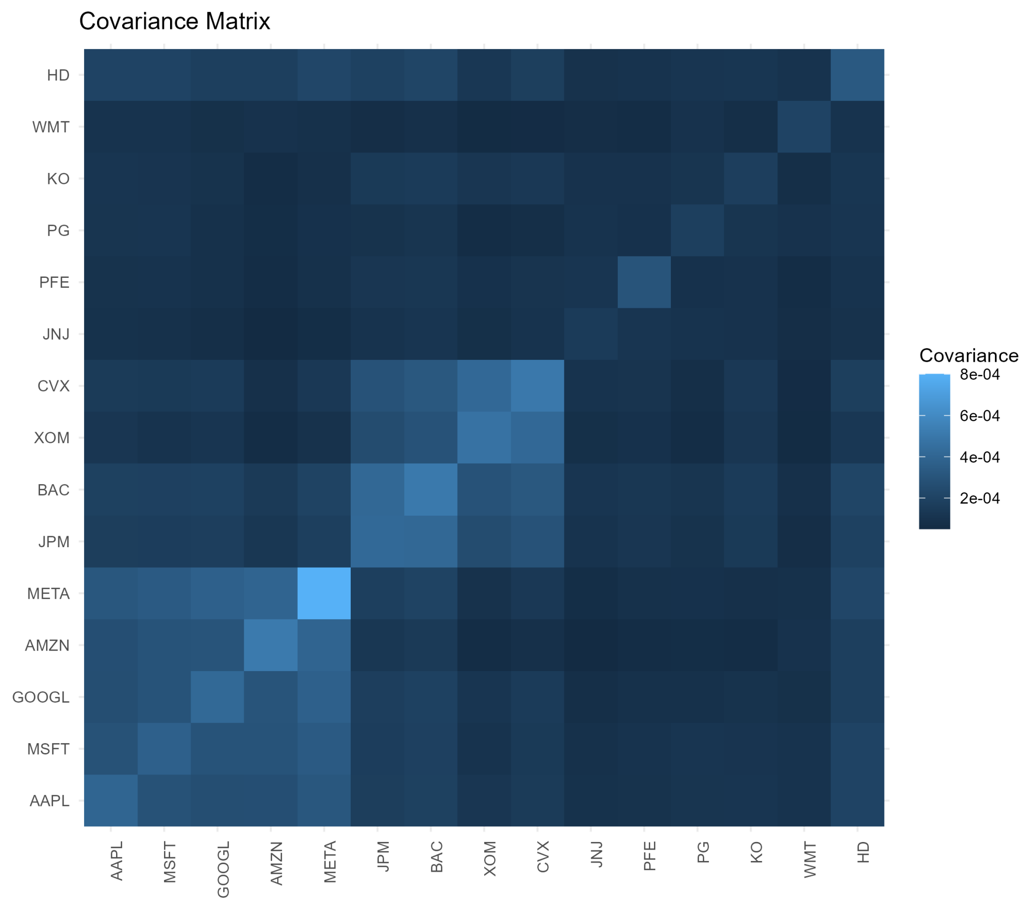

Figure 2.3: Covariance matrix of asset returns, showing the magnitude of joint variability across the selected assets used in portfolio risk estimation and allocation.

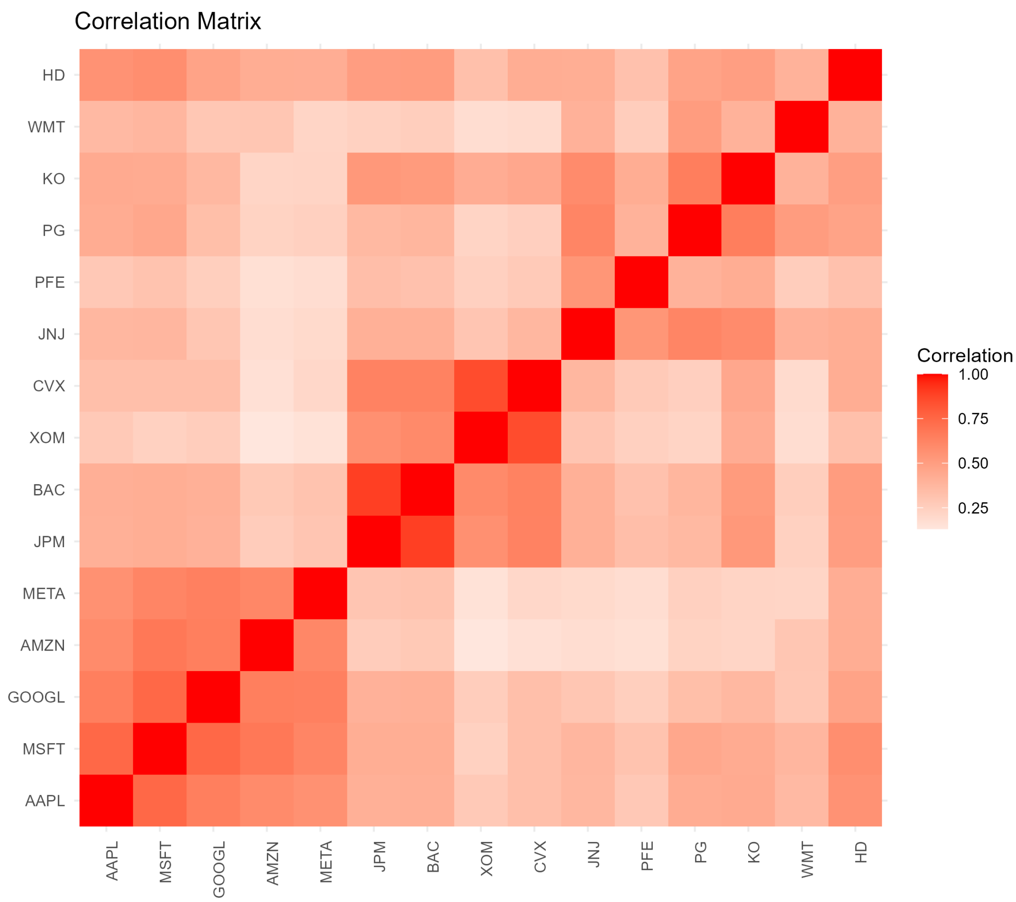

Figure 2.4: Correlation matrix of asset returns, showing the dependence structure across the selected assets prior to hierarchical clustering reordering.

ture of market risk. The decomposition was performed on the covariance matrix of returns, allowing the market structure to be represented in terms of orthogonal components ranked by explained variance.

The explained variance associated with each principal component is summarized in Table 2.2.

Table 2.2: Explained and cumulative variance ratios obtained from Principal Component Analysis (PCA) applied to asset returns.

Component	Explained Variance	Cumulative Variance
PC1	0.4638	0.4638
PC2	0.1793	0.6431
PC3	0.0740	0.7170
PC4	0.0512	0.7682
PC5	0.0466	0.8148
PC6	0.0352	0.8500
PC7	0.0297	0.8798
PC8	0.0262	0.9060
PC9	0.0230	0.9289
PC10	0.0181	0.9470
PC11	0.0144	0.9614
PC12	0.0123	0.9737
PC13	0.0102	0.9839
PC14	0.0084	0.9923
PC15	0.0077	1.0000

The first principal component explains approximately 46% of total variance, while the first two components together account for roughly 64%. The first five components explain more than 80% of total variance, indicating that a relatively small number of latent factors captures much of the variability in the asset universe. Figure 2.5 and Figure 2.6 illustrate this concentration of variance visually.

The first principal component captures a broad market-related factor affecting most assets simultaneously, consistent with its dominant contribution to total variance. Subsequent components explain progressively smaller fractions of variance and reveal more localized sectoral structures that are less visible in the orig-

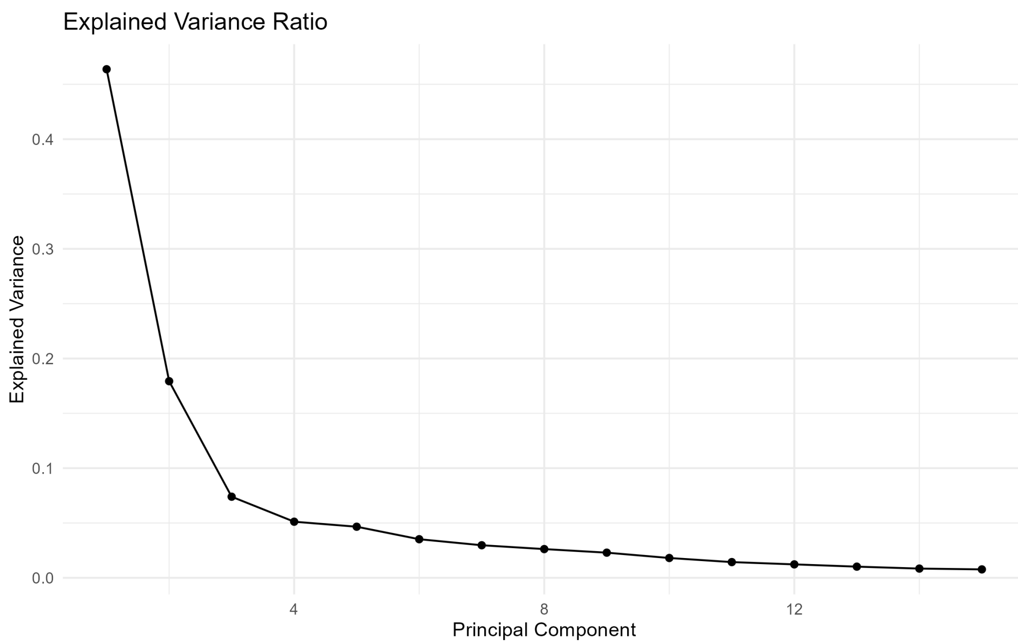

Figure 2.5: Explained variance ratio for each principal component obtained from PCA, showing that the first components capture most of the variability in the asset return space.

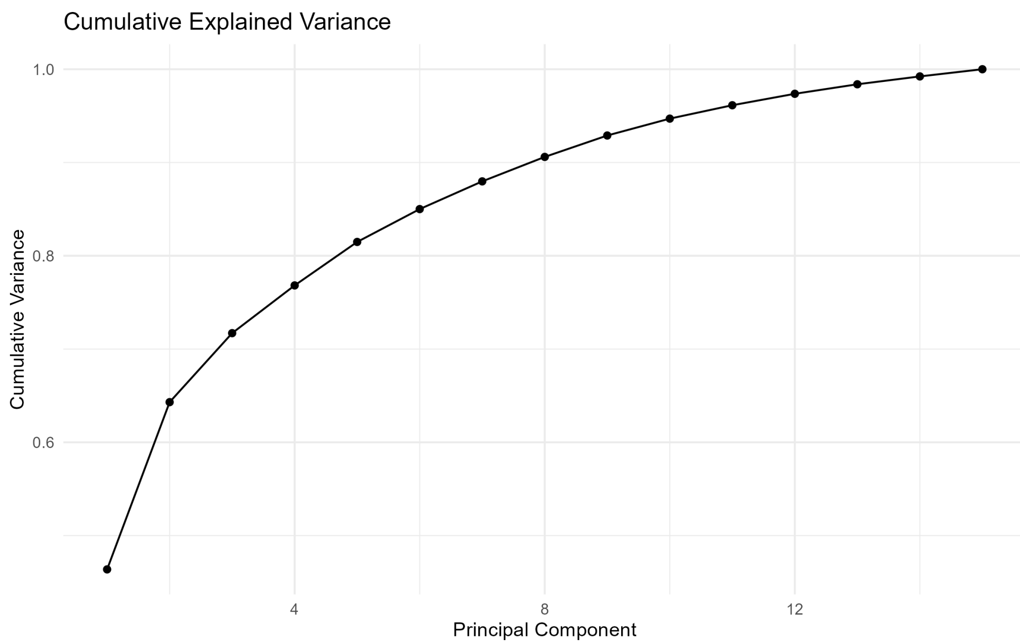

Figure 2.6: Cumulative explained variance obtained from Principal Component Analysis (PCA), showing how the first principal components capture most of the variability in the asset return space.

inal return space.

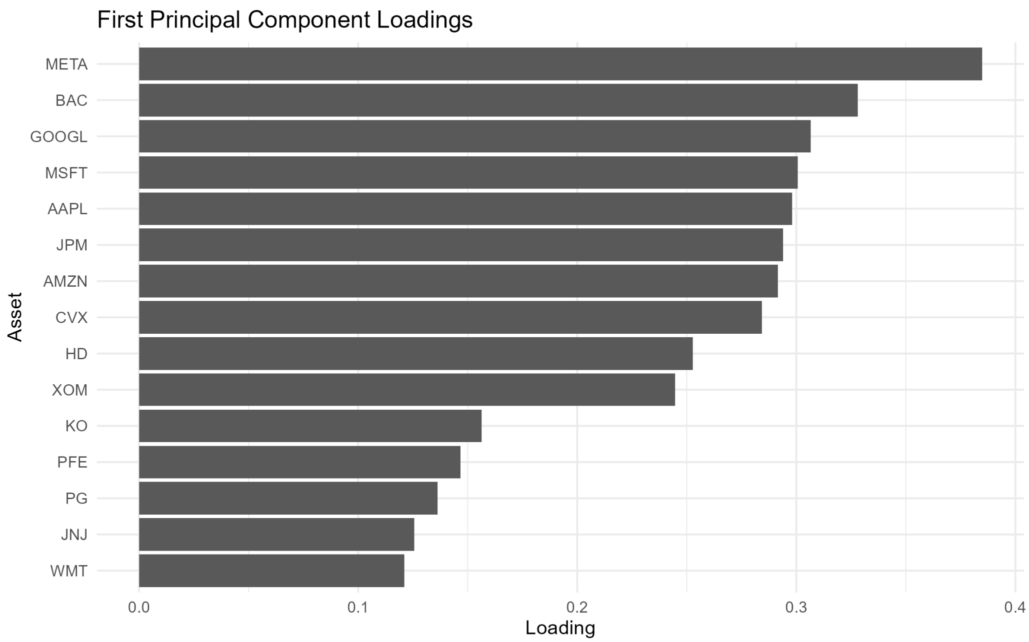

Figure 2.7: Loadings of the first principal component obtained from PCA, illustrating the assets that contribute most strongly to the dominant common risk factor in the portfolio universe.

Following the PCA analysis, hierarchical clustering was applied using the correlation-based distance metric described previously. The clustering procedure organizes assets according to statistical similarity, producing a hierarchical representation of the market structure.

The resulting dendrogram is presented in Figure 2.9.

The hierarchical organization becomes even more evident after reordering the correlation matrix according to the clustering structure, as shown in Figure 2.10.

Several economically intuitive clusters emerge naturally from the data. Technology firms tend to cluster together, energy companies form a separate branch, and defensive sectors exhibit their own hierarchical relationships.

Portfolio construction was then performed using the Hierarchical Risk Parity (HRP) allocation framework. For comparison purposes, an equally weighted portfolio was used as a benchmark throughout the analysis.

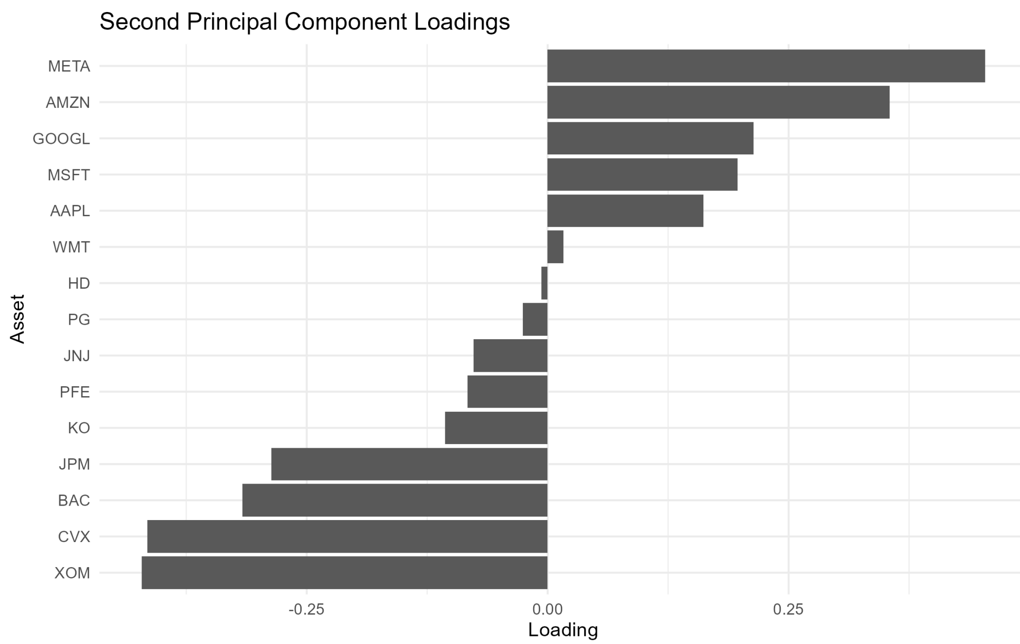

Figure 2.8: Loadings of the second principal component obtained from PCA, highlighting contrasting sectoral exposures and latent diversification patterns within the asset universe.

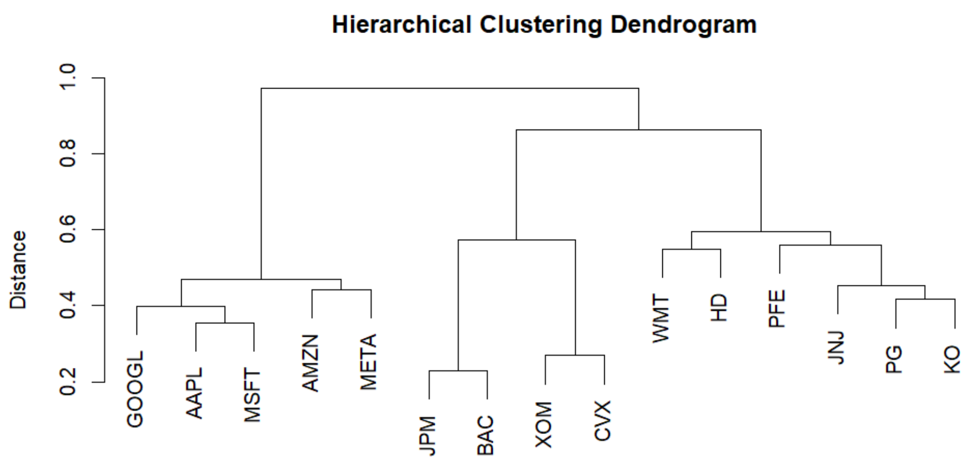

Figure 2.9: Hierarchical clustering dendrogram of the asset universe based on return correlations, revealing sectoral similarity structures used in the Hierarchical Risk Parity (HRP) allocation process.

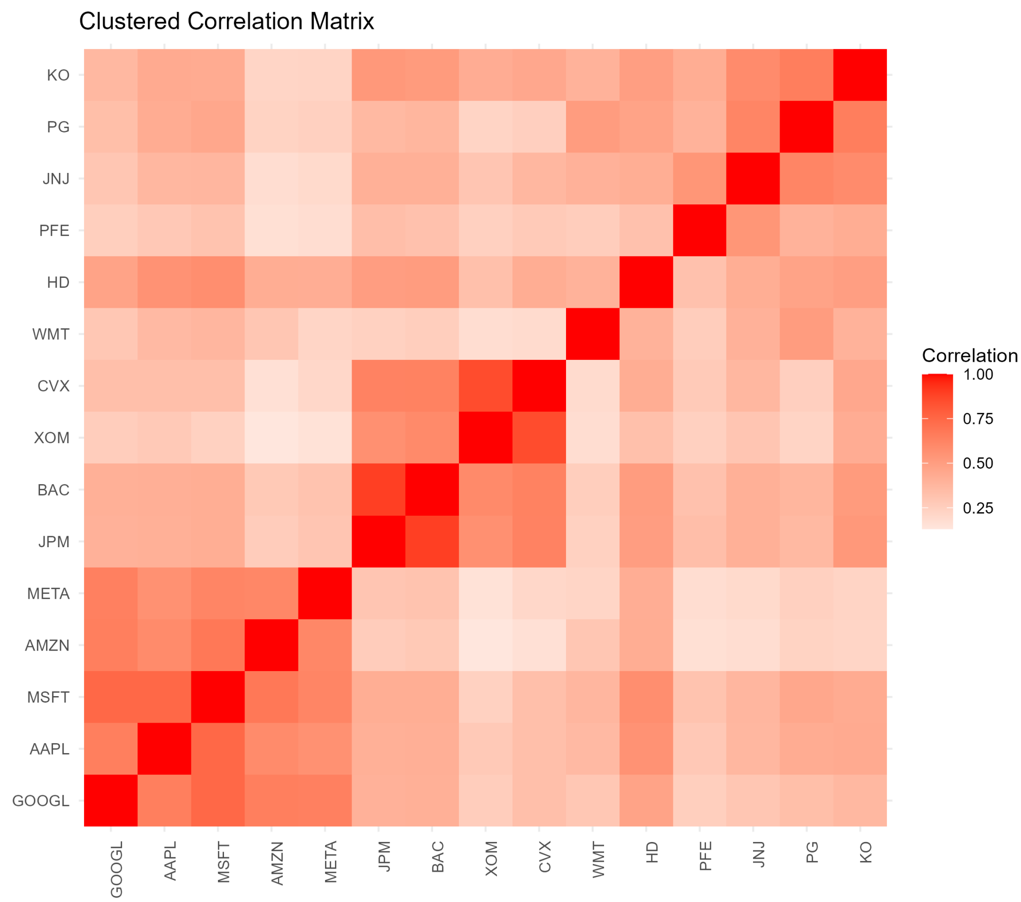

Figure 2.10: Clustered correlation matrix of asset returns, highlighting sectoral structure and similarity patterns used in the hierarchical portfolio construction process.

The equal-weight allocation is shown in Figure 2.11.

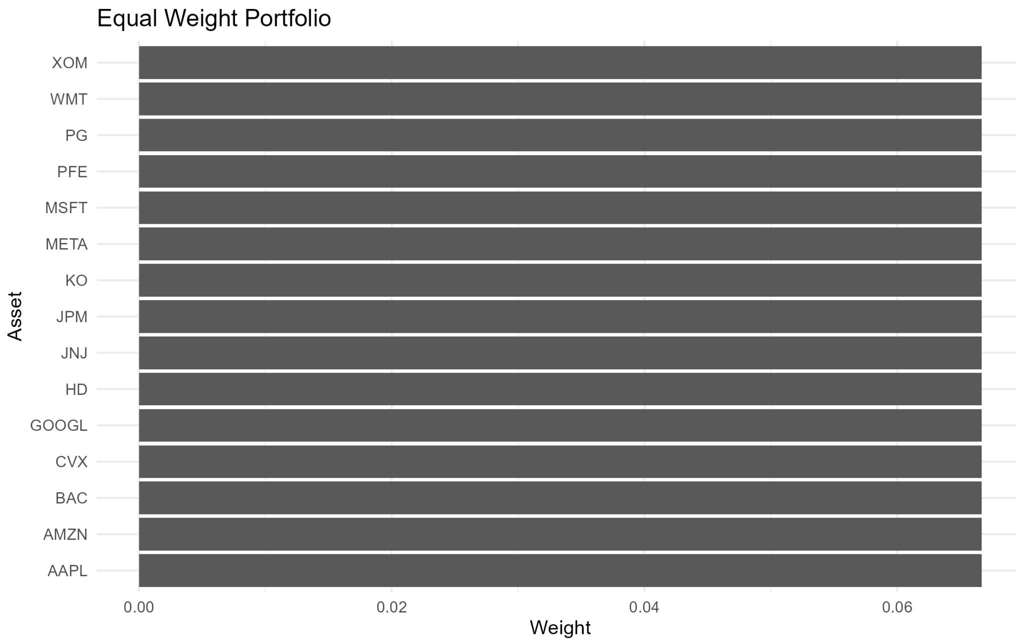

Figure 2.11: Equal-weight portfolio allocation assigning identical weights to all assets in the investment universe.

The HRP allocation produced a substantially different weighting structure. The largest portfolio weights were assigned to JNJ, KO, PG, and WMT, while exposure to several technology, financial, and energy assets was reduced relative to the equal-weight benchmark.

Table 2.3: Final portfolio weights obtained using the Hierarchical Risk Parity (HRP) allocation method.

Asset	Weight
JNJ	0.1285
KO	0.1171
PG	0.1158
WMT	0.1127
GOOGL	0.0705
HD	0.0689

Asset	Weight
PFE	0.0657
JPM	0.0471
MSFT	0.0457
AMZN	0.0439
AAPL	0.0424
BAC	0.0390
XOM	0.0386
CVX	0.0360
META	0.0282

The resulting allocation is visualized in Figure 2.12.

A direct comparison between the equal-weight benchmark and the HRP allocation is shown in Figure 2.13 and Table 2.4.

The differences between the two allocations are visible both at the individual asset level and across sectors. The larger weights assigned to JNJ, KO, PG, and WMT are broadly consistent with the lower volatility and correlation characteristics often associated with defensive sectors.

Table 2.4: Comparison between equal-weight portfolio allocation and Hierarchical Risk Parity (HRP) portfolio weights.

Asset	Equal Weight	HRP
AAPL	0.0667	0.0424
MSFT	0.0667	0.0457
GOOGL	0.0667	0.0705
AMZN	0.0667	0.0439
META	0.0667	0.0282
JPM	0.0667	0.0471
BAC	0.0667	0.0390
XOM	0.0667	0.0386
CVX	0.0667	0.0360
JNJ	0.0667	0.1285
PFE	0.0667	0.0657
PG	0.0667	0.1158

Asset	Equal Weight	HRP
KO	0.0667	0.1171
WMT	0.0667	0.1127
HD	0.0667	0.0689

Risk contributions were also evaluated in order to study how portfolio risk was distributed across assets under the HRP framework.

Table 2.5: Asset-level risk contributions under the Hierarchical Risk Parity (HRP) allocation framework.

Asset	Risk Contribution
KO	0.001158
JNJ	0.001111
PG	0.001090
GOOGL	0.001007
WMT	0.000947
HD	0.000925
JPM	0.000676
MSFT	0.000668
PFE	0.000630
BAC	0.000620
AAPL	0.000615
AMZN	0.000562
CVX	0.000489
XOM	0.000450
META	0.000443

The distribution of risk contributions is illustrated in Figure 2.14.

Portfolio performance was evaluated using cumulative returns, rolling annualized volatility, Sharpe ratio, and maximum drawdown. The cumulative return dynamics of both portfolio strategies are shown in Figure 2.15.

The temporal evolution of portfolio risk is shown in Figure 2.16.

Drawdown dynamics were analyzed separately in order to study the behaviour of both portfolios during periods of market stress.

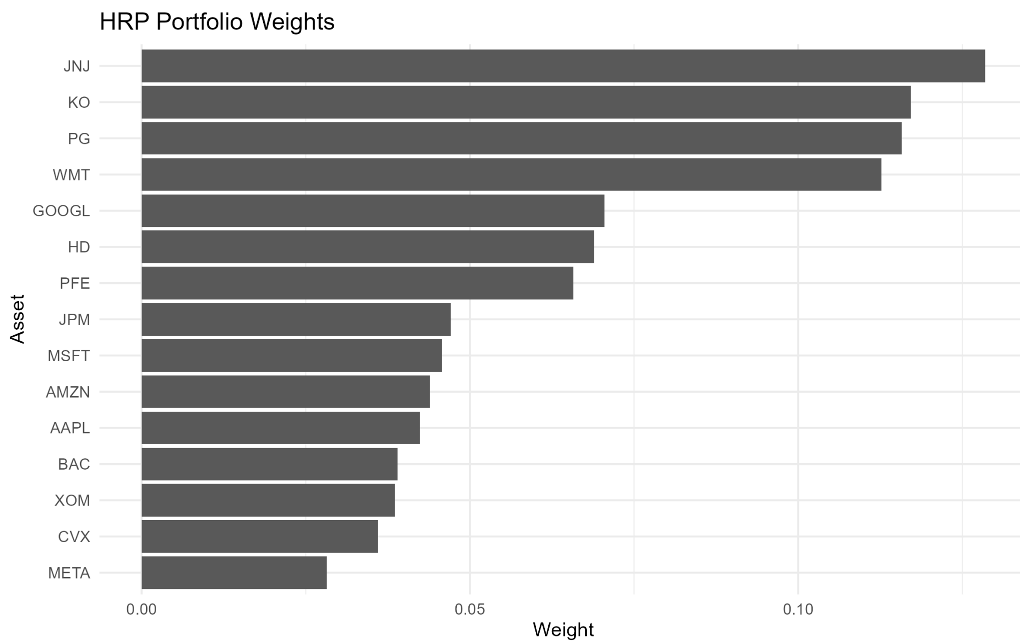

Figure 2.12: Portfolio weights obtained using the Hierarchical Risk Parity (HRP) allocation method, reflecting the hierarchical correlation structure and risk diversification across assets.

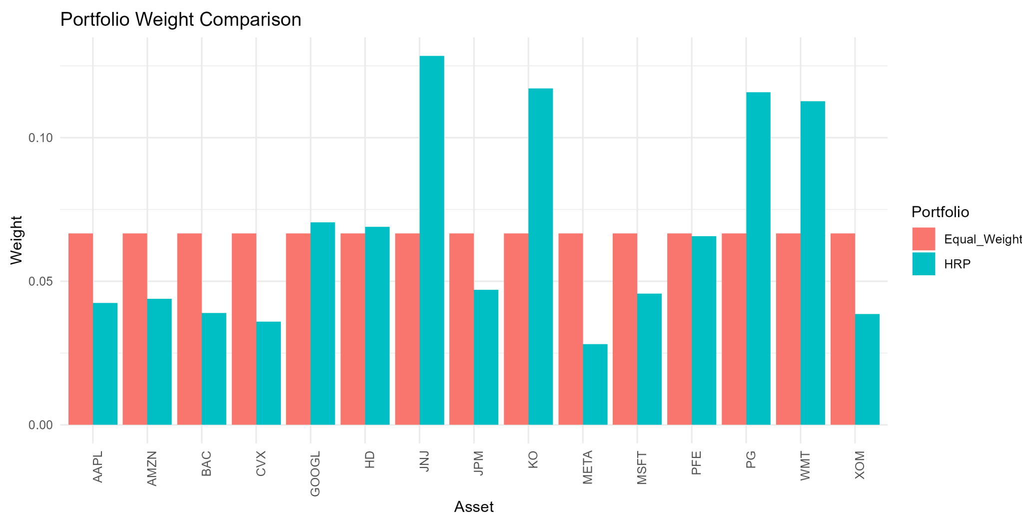

Figure 2.13: Comparison between equal-weight and Hierarchical Risk Parity (HRP) portfolio allocations, highlighting how the HRP framework redistributes exposure according to the hierarchical correlation and risk structure of the asset universe.

The final performance summary is presented in Table 2.6.

Table 2.6: Performance comparison between the Hierarchical Risk Parity (HRP) portfolio and the equal-weight benchmark portfolio.

Portfolio	Annualized Return	Annualized Volatility	Sharpe Ratio	Maximum Drawdown
HRP Portfolio	0.1754	0.2021	0.8680	-0.3139
Equal Weight Portfolio	0.1514	0.1808	0.8373	-0.2831

The HRP portfolio achieved a modest improvement in annualized return and Sharpe ratio relative to the equal-weight benchmark. However, this improvement was accompanied by a somewhat deeper maximum drawdown during the evaluation period, illustrating that diversification benefits do not necessarily translate into lower losses under all market conditions.

The resulting workflow combines PCA, hierarchical clustering, and risk-based

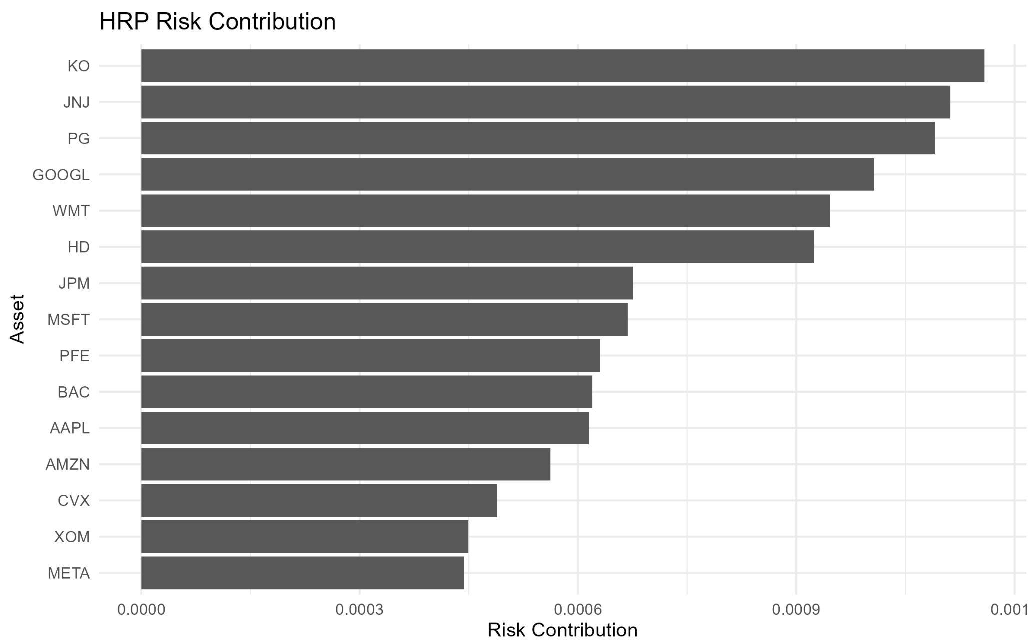

Figure 2.14: Risk contribution of each asset under the Hierarchical Risk Parity (HRP) allocation, illustrating how portfolio risk is distributed across the investment universe.

Figure 2.15: Cumulative returns of the compared portfolio strategies over the evaluation period, illustrating differences in long-term growth dynamics and drawdown behavior.

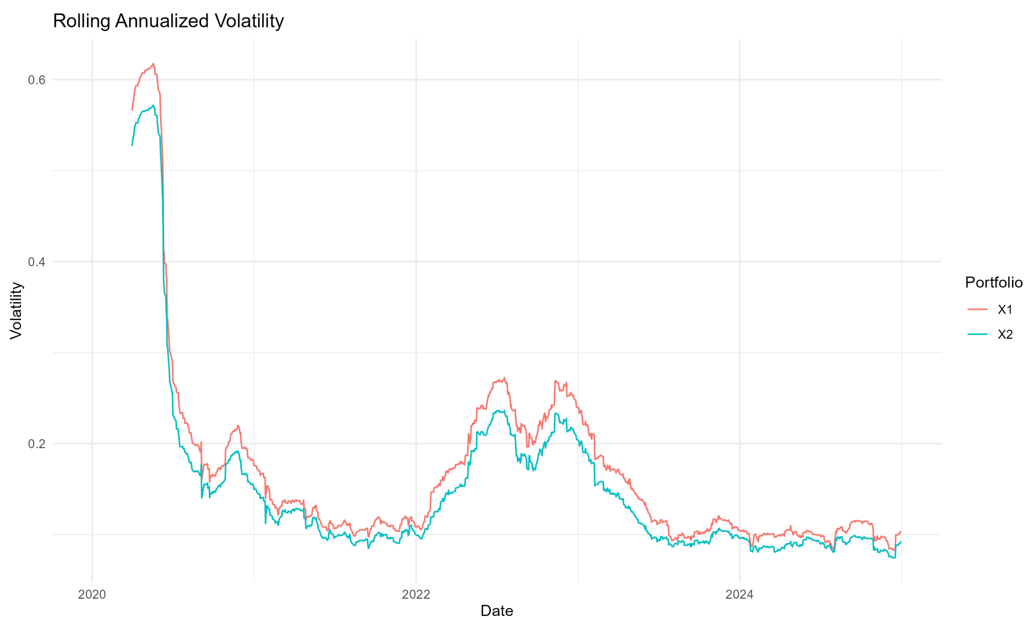

Figure 2.16: Rolling annualized volatility of the portfolio strategies over time, illustrating the dynamic evolution of portfolio risk under different allocation frameworks.

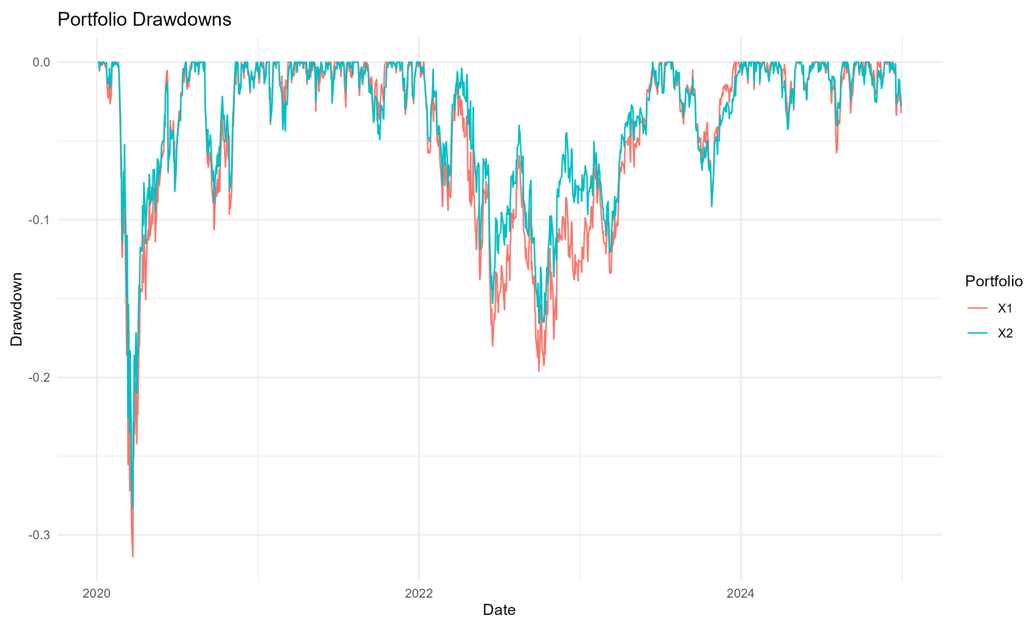

Figure 2.17: Portfolio drawdowns over the evaluation period, comparing the magnitude and persistence of temporary losses across the portfolio strategies.

portfolio allocation to study diversification from a structural rather than purely optimization-based perspective. Rather than focusing exclusively on return maximization, the analysis emphasizes robustness, interpretability, and stability under realistic market conditions.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

2.4 Results, Discussion, and Limitations

The results of the analysis suggest that the selected asset universe exhibits a considerably more structured organization of risk than might be inferred from asset labels or sector classifications alone. Although the portfolio contains companies operating across multiple industries, both the PCA decomposition and the hierarchical clustering analysis indicate that a relatively small number of latent factors account for a large proportion of observed market variability.

The PCA results provide the clearest evidence of this concentration. The first principal component alone explains approximately 46% of total variance, while the first two components account for roughly 64%. More than 80% of total variance is captured by the first five components. These findings suggest that much of the apparent diversity within the asset universe is driven by a limited set of common risk sources. From a portfolio construction perspective, this highlights an important distinction between holding many assets and achieving genuine diversification. Increasing the number of securities does not necessarily increase the number of independent sources of risk.

The principal component loadings further support this interpretation. The first component exhibits broad exposure across most assets and behaves consistently with a dominant market factor affecting the portfolio universe as a whole. Subsequent components reveal more localized patterns associated with sector-specific behaviour and relative differences between groups of assets. Together, these results illustrate how PCA can provide a compact representation of market structure while helping identify hidden concentrations of risk that may not be immediately visible in the original return space.

A complementary perspective emerges from the hierarchical clustering analysis. The dendrogram and clustered correlation matrix reveal economically intuitive groupings that arise directly from the data. Technology companies tend to cluster together, energy firms form a separate branch, and defensive sectors exhibit their own hierarchical relationships. Rather than imposing predefined classifications, the clustering procedure allows the statistical organization of the market to emerge from observed return behaviour. This provides a useful framework for understanding how diversification is distributed across the portfolio and where concentrations of risk may exist.

The portfolio construction stage extends these observations into an allocation framework. Compared with the equal-weight benchmark, the Hierarchical Risk Parity portfolio produces a noticeably different allocation structure. The largest weights were assigned to JNJ, KO, PG, and WMT, while exposure to several technology, financial, and energy assets was reduced. Importantly, this outcome should not be interpreted as a prediction that these assets will outperform in the future. Rather, it reflects their role within the correlation structure of the portfolio and the objective of distributing risk more evenly across the hierarchy identified during the clustering stage.

This distinction is central to the interpretation of the results. The primary contribution of HRP is not necessarily superior return generation, but a different approach to diversification. Traditional portfolio construction methods often allocate capital directly, whereas HRP attempts to allocate risk. As a consequence, two portfolios with similar levels of capital diversification may exhibit substantially different risk concentrations. The risk contribution analysis suggests that the HRP framework achieves a more balanced distribution of portfolio risk across assets, consistent with its underlying design objectives.

Performance comparisons between the HRP portfolio and the equal-weight benchmark reveal both strengths and trade-offs. The HRP portfolio achieved a modest improvement in annualized return and Sharpe ratio relative to the benchmark. At the same time, it experienced a somewhat deeper maximum drawdown during the evaluation period. These results illustrate an important practical point: improvements in one dimension of portfolio performance do not necessarily translate into improvements across all dimensions. Diversification, risk balancing, and return generation remain competing objectives, and no allocation methodology can fully eliminate these trade-offs.

The results should therefore be interpreted as evidence that unsupervised learning techniques can provide useful insights into market structure and portfolio construction rather than as proof of the superiority of a particular allocation strategy. The analysis demonstrates that PCA and hierarchical clustering are capable of uncovering meaningful relationships within the asset universe and that these relationships can be incorporated into a systematic allocation process. Whether this ultimately leads to superior long-term investment performance remains dependent on market conditions, investment horizons, transaction costs, and implementation choices.

Several limitations should also be acknowledged. First, the analysis was conducted using a relatively small universe of fifteen assets. While sufficient for illustrating the methodology, larger investment universes may exhibit more complex correlation structures and different clustering behaviour. Second, the study relies on a single historical sample period. Financial markets are inherently dynamic, and correlation structures evolve through time, particularly during periods of economic stress. Consequently, both PCA decompositions and clustering results may vary across market regimes.

A further limitation is that the analysis assumes frictionless portfolio implementation. Transaction costs, taxes, liquidity constraints, and rebalancing considerations were not incorporated into the evaluation. In practice, these factors can materially influence realized portfolio performance and may reduce some of the theoretical benefits of more sophisticated allocation methodologies.

Finally, although HRP was designed to improve robustness relative to covariance-sensitive optimization frameworks, it does not eliminate market risk or guarantee protection during severe market downturns. Common risk factors can still dominate portfolio behaviour when correlations rise across the market, reducing the effectiveness of diversification precisely when it is needed most.

Taken together, the results suggest that unsupervised learning techniques provide a valuable lens through which to study diversification and market structure. The main contribution of PCA and HRP in this context is not the pursuit of maximum historical return, but the ability to make sources of risk more explicit and to organize portfolio construction around the underlying structure of the market. From this perspective, the project illustrates how statistical learning methods can complement traditional portfolio theory by offering a more interpretable and structurally informed approach to asset allocation.

2.5 Technical Appendix (Optional)

This appendix provides supplementary mathematical and methodological details related to the unsupervised learning techniques used throughout the project. Additional background on PCA and clustering can be found in [9] and [15].

The objective is not to present a comprehensive treatment of the underlying methods, but rather to summarize the concepts most relevant to the portfolio construction process. The material presented here supports the interpretation of the results and clarifies some of the modeling choices discussed in the main text.

The appendix is organized into two sections. The first summarizes the principal concepts behind Principal Component Analysis (PCA), which was used to study the structure of asset return variability and identify dominant sources of common variation. The second discusses Hierarchical Clustering (HC), which was used to group assets according to their similarity patterns and support diversification decisions.

The main body of the project is intended to remain fully understandable without consulting this appendix. Readers interested primarily in the practical application and results may safely skip the material that follows.

2.5.1 Principal Component Analysis (PCA)

2.5.1.1 Centering of Returns

Let

$$X = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{n1} & r_{n2} & \cdots & r_{np} \end{bmatrix} \tag{2.4}$$

denote the matrix of asset returns, where each column corresponds to an asset and each row represents an observation period.

Prior to applying PCA, the return series were centered by subtracting the sample mean of each asset:

$$x_{ij}^{(c)} = r_{ij} - \bar{r}_j \tag{2.5}$$

where $\bar{r}_j$ denotes the average return of asset j.

Centering removes differences in mean return levels and ensures that the subsequent analysis focuses exclusively on the covariance structure of the data. The centered return matrix was then used to construct the covariance matrix and perform the eigendecomposition underlying PCA.

2.5.1.2 Covariance Matrix

The covariance matrix summarizes the dependence structure among asset returns and is given by

$$\Sigma = \frac{1}{n-1} X_c^\top X_c \tag{2.6}$$

where X_c denotes the centered return matrix and n is the number of observations.

Each element σ_{ij} measures the extent to which the returns of assets i and j vary together. Positive values indicate that returns tend to move in the same direction, whereas negative values indicate opposite movements.

Because PCA is based on the covariance structure of the data, the matrix Σ serves as the primary object of analysis. The principal components are obtained by identifying the directions that explain the largest share of the variability captured by this matrix.

2.5.1.3 Eigenvalue Decomposition

PCA is based on the eigendecomposition of the covariance matrix,

$$\Sigma = V \Lambda V^{\top} \tag{2.7}$$

where V is the matrix of eigenvectors and Λ is a diagonal matrix containing the corresponding eigenvalues.

Each eigenvector defines a principal component, while the associated eigenvalue measures the amount of variance explained by that component.

The principal components are mutually orthogonal and provide a new coordinate system in which the dominant patterns of variation become easier to identify. Components associated with larger eigenvalues explain a greater share of total variance and therefore contain more information about the structure of asset returns.

In this project, the eigendecomposition of the covariance matrix provided the basis for evaluating how much of the variability in the investment universe could be explained by a reduced number of components.

2.5.1.4 Explained Variance

The eigenvalues obtained from the covariance matrix quantify the amount of variance explained by each principal component. Let λ_i denote the eigenvalue associated with the i-th principal component. The proportion of variance explained by that component is given by

$$\mathrm{PVE}_i = \frac{\lambda_i}{\sum_{j=1}^{p} \lambda_j} \tag{2.8}$$

where p denotes the total number of assets.

The cumulative proportion of explained variance can be expressed as

$$\mathrm{CPVE}_k = \sum_{i=1}^{k} \mathrm{PVE}_i \tag{2.9}$$

representing the fraction of total variance retained when only the first k principal components are considered.

The explained variance analysis indicated a substantial concentration of variability in a small number of components. The first principal component accounted for approximately 46% of total variance, while the first two components explained roughly 64%. The first five components captured more than 80% of the variability observed across the investment universe.

These results suggest that a relatively small number of latent factors drive a large fraction of asset return behavior. Consequently, the effective dimensionality of the dataset is considerably smaller than the number of assets included in the analysis.

2.5.1.5 Principal Component Loadings

While explained variance quantifies the importance of each principal component, the component loadings describe how individual assets contribute to those components.

Let

$$v_k = \begin{bmatrix} v_{1k} \\ v_{2k} \\ \vdots \\ v_{pk} \end{bmatrix} \tag{2.10}$$

denote the eigenvector associated with the k-th principal component. The elements of this vector are commonly referred to as loadings.

Assets with large absolute loadings contribute more strongly to the corresponding component, whereas assets with loadings closer to zero have a smaller influence on that source of variation.

The loadings associated with the first two principal components were examined to identify the assets contributing most strongly to the dominant patterns of return variability. Because principal components are linear combinations of the original assets, the loading structure provides insight into how different securities participate in the latent factors revealed by the PCA decomposition.

Together with the explained variance analysis, the loadings help characterize the underlying structure of the investment universe and identify groups of assets that exhibit similar exposures to common sources of variation.

2.5.1.6 Interpretation for Asset Allocation

The PCA results indicate that a substantial portion of total variance can be explained by a relatively small number of principal components. From a portfolio construction perspective, this suggests that many assets are influenced by common underlying factors rather than behaving independently.

As a consequence, the number of securities held in a portfolio may overstate the true degree of diversification achieved. Assets exposed to the same dominant components can contribute similar sources of risk even when they belong to different sectors or industries.

For this reason, PCA can be viewed as a diagnostic tool for assessing the structure of risk within an investment universe. Rather than focusing on individual assets, the analysis emphasizes the latent factors that drive collective movements in returns.

In this project, PCA was used primarily as an exploratory technique to investigate the dimensionality of the dataset and to identify common sources of variation prior to the clustering analysis described in the following section.

2.5.2 Hierarchical Clustering (HC)

2.5.2.1 Correlation-Based Distance

The clustering analysis was based on the correlation structure of asset returns. Let ρ_{ij} denote the Pearson correlation coefficient between assets i and j. The corresponding distance measure is defined as

$$d_{ij} = \sqrt{\frac{1-\rho_{ij}}{2}} \tag{2.11}$$

This transformation converts correlations into distances while preserving the properties required by a distance metric.

Several useful relationships follow directly from this definition:

- If $\rho_{ij} = 1$, then $d_{ij} = 0$.
- If $\rho_{ij} = 0$, then $d_{ij} = \sqrt{1/2}$.
- If $\rho_{ij} = -1$, then $d_{ij} = 1$.

Assets exhibiting similar return dynamics therefore appear close to one another in the resulting distance matrix, whereas assets with dissimilar behavior appear farther apart.

The correlation-based distance matrix served as the primary input to the hierarchical clustering procedure implemented in this project.

2.5.2.2 Agglomerative Clustering

Hierarchical clustering was performed using an agglomerative approach.

The procedure begins by assigning each asset to its own cluster. At each iteration, the two clusters exhibiting the smallest distance are merged into a new cluster. This process continues until all assets have been combined into a single hierarchical structure.

If the investment universe contains p assets, the algorithm initially creates p individual clusters. Each merge reduces the number of clusters by one until the hierarchy is complete.

The resulting sequence of merges reveals how assets are related at different levels of similarity. Assets that merge early in the process tend to exhibit highly similar return patterns, whereas assets that merge only at later stages are generally less related.

2.5.2.3 Ward's Linkage Method

After computing the distance matrix, the clustering algorithm must determine how distances between clusters are evaluated following each merge. This choice is known as the linkage criterion.

In this project, Ward's linkage method was employed through the Ward.D2 algorithm.

Ward's method seeks to minimize the increase in within-cluster variance produced by each merge. At every iteration, the algorithm selects the pair of clusters whose combination results in the smallest increase in overall cluster heterogeneity.

This approach tends to generate compact and relatively balanced clusters, making it particularly useful for identifying coherent groups of assets exhibiting similar return behavior.

2.5.2.4 Dendrogram Interpretation

The hierarchical structure produced by the clustering algorithm can be visualized through a dendrogram.

A dendrogram is a tree-like representation that displays the sequence of cluster merges performed during the agglomerative procedure. Each terminal node corresponds to an individual asset, while internal nodes represent cluster combinations formed throughout the algorithm.

The vertical axis represents the distance at which clusters are merged. Assets connected at lower heights tend to exhibit stronger similarity, whereas assets that merge only at higher levels are generally less related.

The dendrogram presented in the main body of the project reveals the hierarchical organization of the investment universe and highlights groups of assets exhibiting similar return behavior.

2.5.2.5 Clustered Correlation Matrix

The ordering produced by hierarchical clustering can be used to reorganize the correlation matrix according to the cluster structure identified in the data.

Rather than displaying assets in an arbitrary order, the reordered matrix places statistically similar assets next to one another. As a result, groups of highly correlated assets appear as visible blocks along the diagonal of the matrix.

This representation provides a convenient way to inspect the dependency structure of the investment universe and to identify groups of assets sharing similar return dynamics.

In this project, the clustered correlation matrix complemented the dendrogram by providing an alternative visualization of the relationships among assets.

2.5.3 Diversification Through Clustering

The clustering results indicate that similarities among assets are not distributed uniformly throughout the investment universe. Instead, groups of assets exhibiting comparable return behavior emerge naturally from the data.

From a portfolio construction perspective, these clusters can reveal hidden concentrations of risk. A portfolio may contain a large number of securities while remaining exposed to a relatively small set of common drivers if many holdings belong to the same cluster.

Hierarchical clustering provides a complementary perspective on diversification by focusing on relationships among assets rather than on individual securities. Assets belonging to different clusters are more likely to contribute distinct sources of risk and therefore may improve portfolio diversification.

Together, PCA and hierarchical clustering provide complementary views of the structure of asset returns. While PCA identifies dominant sources of variation, clustering reveals how assets are organized according to similarity patterns within the investment universe.

2.5.3.1 Methodological Limitations

PCA captures linear relationships among asset returns and may not fully represent nonlinear dependency structures present in financial markets.

Similarly, the clustering structure obtained depends on both the distance metric and the linkage criterion employed. Alternative specifications may produce different cluster configurations and lead to different interpretations of the investment universe.

Despite these limitations, PCA and hierarchical clustering remain useful exploratory tools for understanding the structure of asset returns and supporting diversification analysis.

Chapter 3

Option Pricing with Path Integrals

Most option pricing models focus on the value of a contract at maturity. Path integral methods approach the same problem from a different perspective: instead of analysing a single outcome, they evaluate the contribution of all possible price trajectories that could lead to it. This viewpoint is particularly appealing for derivatives whose value depends not only on the final price of the underlying asset, but also on the path followed to reach it.

While the Black–Scholes framework remains one of the most influential models in quantitative finance, many derivatives require numerical techniques because closed-form solutions are either unavailable or difficult to obtain. Path integral methods provide an alternative formulation that naturally expresses option values as weighted averages over entire trajectories, creating a direct connection between option pricing and simulation-based methods.

This project investigates whether a path integral formulation can serve as a practical computational framework for option pricing. The analysis focuses on European call options, where analytical Black–Scholes prices are available and can be used as a benchmark. A path integral representation of the pricing problem is derived, implemented numerically using the Metropolis–Hastings algorithm, and evaluated against both the analytical Black–Scholes solution and conventional Monte Carlo estimates. Historical market data is incorporated to illustrate how the methodology can be applied in a realistic setting.

3.1 Problem Definition

For a standard European option, the Black–Scholes model provides an elegant closed-form solution under a set of simplifying assumptions. The situation becomes more challenging when dealing with contracts that involve early exercise features, path dependence, or more complex market dynamics. In these cases, analytical solutions may no longer exist, and numerical methods become essential.

Most numerical approaches are built around either solving a pricing equation or approximating the evolution of the underlying asset through discrete structures such as lattices. Path integral methods offer a different perspective. Rather than focusing on a single terminal outcome, they formulate the valuation problem as an aggregation over all possible trajectories that the underlying asset could follow between the present and maturity.

This trajectory-based viewpoint is particularly attractive for derivatives whose value depends on the history of the underlying asset rather than on its final price alone. It also provides a natural bridge between option pricing and simulation-based techniques, allowing the valuation problem to be reformulated in terms of sampling and aggregating large ensembles of possible price paths.

The central question explored in this project is whether this alternative formulation can provide a practical foundation for option pricing while remaining consistent with established financial theory. The goal is not to outperform Black–Scholes where an analytical solution already exists, but to examine whether path integrals can serve as a flexible framework for pricing problems where closed-form solutions are no longer available.

3.2 Modeling Approach

The starting point of the model is the same stochastic process used in the Black–Scholes framework. The underlying asset is assumed to follow a geometric Brownian motion, allowing the option pricing problem to be expressed in terms of the probability distribution of future price trajectories.

Rather than working directly with the Black–Scholes partial differential equation, the pricing problem is reformulated using a path integral representation. In this formulation, the value of an option is obtained by aggregating the contribution of all admissible trajectories connecting the current asset price to a future state at maturity. Each trajectory is assigned a statistical weight determined by the underlying stochastic process and contributes to the expected discounted payoff.

To make the problem computationally tractable, the continuous time horizon is discretized into a finite number of intervals. This transforms the original path integral into a multidimensional integration problem whose dimensionality grows with the number of time steps used to represent the trajectory. The resulting formulation provides a numerical approximation of the Black–Scholes propagator while preserving the probabilistic interpretation of the original model.

Evaluating these multidimensional integrals directly quickly becomes impractical as the number of dimensions increases. For this reason, the numerical estimation is performed using Markov Chain Monte Carlo techniques. Among the available sampling methods, the Metropolis–Hastings algorithm was selected because it allows representative trajectories to be generated efficiently from the target distribution without requiring exhaustive exploration of the integration domain.

The analysis focuses on European call options, where analytical Black–Scholes prices are available and can be used as a benchmark. This provides a controlled setting in which the numerical estimates obtained from the path integral formulation can be compared against a well-established analytical solution before considering more complex derivatives.

The overall modeling strategy combines three components: a geometric Brownian motion describing the evolution of the underlying asset, a path integral formulation that expresses option values as weighted averages over entire price trajectories, and a Markov Chain Monte Carlo algorithm used to estimate those values numerically.

3.3 Methodology and Implementation

The objective of this project is not to compete with established option pricing methods, but to explore how ideas originating from the path integral formulation can be translated into a practical numerical workflow. To keep the analysis transparent, the implementation is organized around two experiments. The first uses a controlled setting in which the analytical solution is known. The second applies the same workflow to observed market data.

3.3.1 Controlled Experiment

The analysis begins with a European call option under the Black-Scholes assumptions. The parameters used throughout the baseline experiment are summarized in Table 3.1.

Table 3.1: Model parameters used in the baseline pricing experiment.

Parameter	Value
Initial Asset Price	100.0000
Strike Price	100.0000
Time to Maturity	1.0000
Risk-Free Rate	0.0500
Volatility	0.2000

Under these assumptions, the Black-Scholes formula provides an exact analytical price. This creates a useful benchmark against which numerical estimators can be evaluated.

Although the path integral formulation is naturally expressed in terms of entire asset price trajectories, a European call option depends only on the terminal asset price. For this reason, the implementation does not attempt to sample complete paths through Markov Chain Monte Carlo. Instead, the problem is simplified by sampling from the terminal log-price distribution implied by the risk-neutral geometric Brownian motion model. This reduction preserves the essential pricing problem while keeping the numerical implementation manageable.

Simulated price paths are nevertheless generated to illustrate the stochastic dynamics underlying the model and to provide a visual representation of the path-based perspective motivating the analysis.

Three pricing approaches are then compared. The Black-Scholes formula serves as the analytical benchmark, standard Monte Carlo simulation provides a conventional numerical estimator, and a Metropolis-Hastings sampler is used to generate samples from the terminal risk-neutral distribution.

The resulting estimates are reported in Table 3.2 and visualized in Figure 3.2.

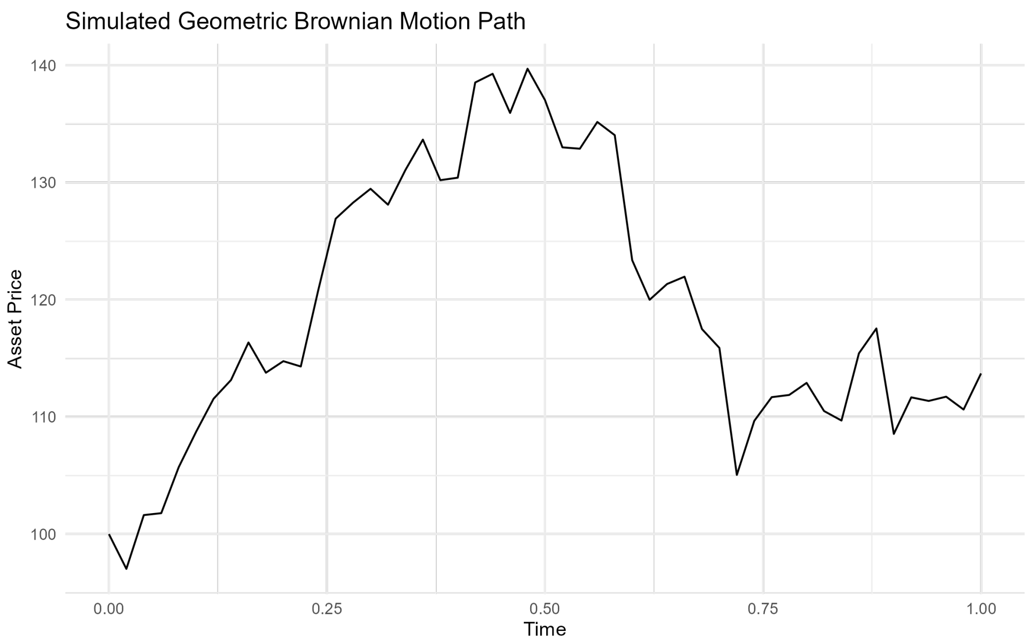

Figure 3.1: A subset of simulated asset-price trajectories under the risk-neutral geometric Brownian motion model. Although the final implementation estimates option values from the terminal distribution, the simulated paths provide a visual representation of the path-based framework motivating the analysis.

Table 3.2: Comparison of option prices obtained using Black-Scholes, Monte Carlo simulation, and Metropolis-Hastings sampling.

Method	Option Price	Absolute Error
Black-Scholes	10.4506	0.0000
Monte Carlo	10.3896	0.0610
Metropolis-Hastings	10.5800	0.1295

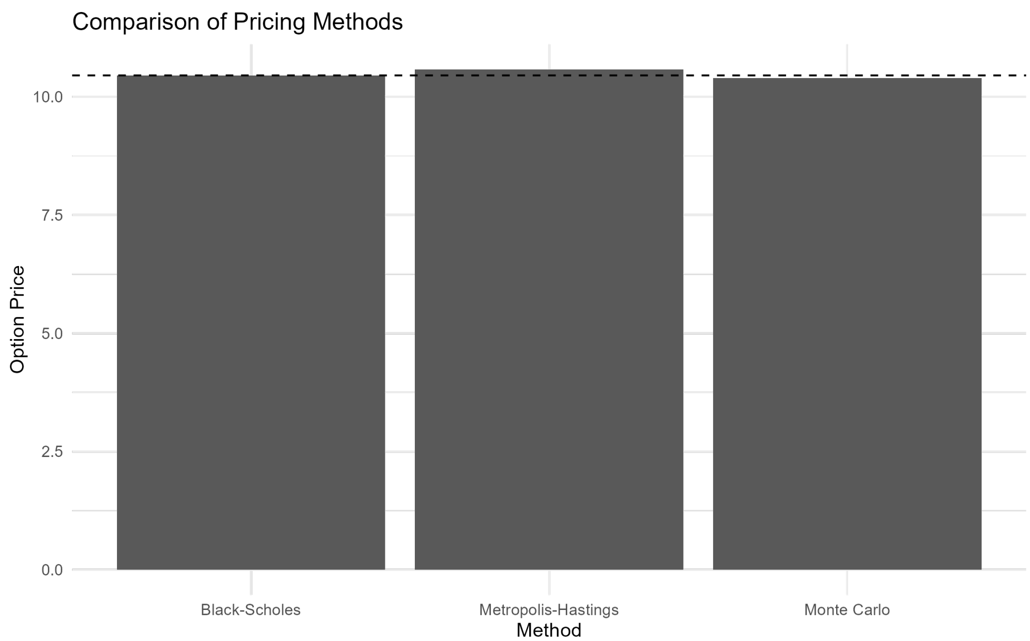

Figure 3.2: Comparison of option prices obtained using analytical and simulation-based methods.

All three methods produce estimates close to the analytical benchmark. In this simple setting, standard Monte Carlo achieves the smallest numerical error. This result is not surprising. For a European call option, direct Monte Carlo sampling is already an efficient estimator, and the additional complexity introduced by Markov Chain Monte Carlo does not provide a clear advantage.

To better understand the behavior of the numerical estimators, the experiment is repeated for different sample sizes. The results are shown in Table 3.3 and Figure 3.3.

Table 3.3: Convergence behavior of Monte Carlo and Metropolis-Hastings estimators as the number of samples increases.

Samples	Monte Carlo	Metropolis-Hastings	Black-Scholes
100	9.2822	8.3345	10.4506
500	11.4302	9.4567	10.4506
1000	10.0559	11.6238	10.4506
5000	10.3909	9.8939	10.4506
10000	10.4008	10.8390	10.4506

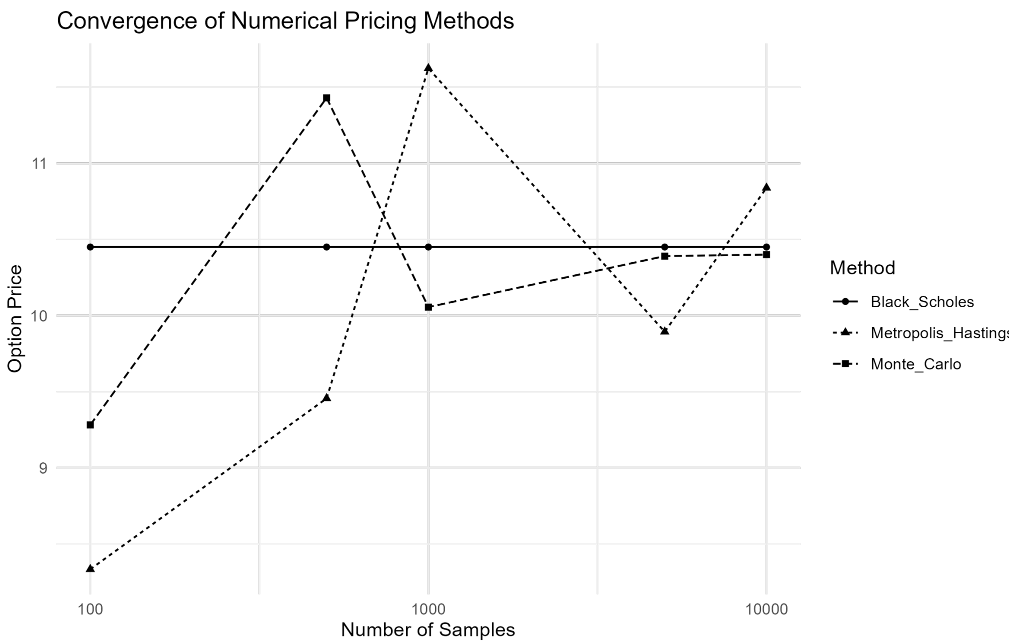

Figure 3.3: Convergence of Monte Carlo and Metropolis-Hastings estimates toward the Black-Scholes benchmark as the sample size increases.

Both estimators move toward the analytical solution as the sample size increases. The Monte Carlo estimator exhibits a more stable convergence pattern, while the Metropolis-Hastings estimates show larger fluctuations at smaller sample sizes. This behavior is consistent with the autocorrelation introduced by Markov chain sampling and illustrates one of the practical trade-offs associated with MCMC-based methods.

The purpose of this comparison is not to demonstrate that Metropolis-Hastings outperforms Monte Carlo in a problem where the analytical solution is already available. Rather, it provides a controlled environment for validating the implementation and understanding the numerical behavior of the estimator before considering more complex derivatives.

3.3.2 Market Data Example

After validating the workflow in a controlled setting, the same methodology is applied using observed market data. Historical prices for AAPL are downloaded from Yahoo Finance and used to estimate a spot price and historical volatility.

Figure 3.4: Historical AAPL price series used in the market-data example.

The resulting inputs are summarized in Table 3.4.

Table 3.4: Market-derived inputs obtained from historical AAPL data.

Parameter	Value
Ticker	AAPL

Parameter	Value
Spot Price	190.3751
Strike Price	190.3751
Time to Maturity	1
Risk-Free Rate	0.05
Historical Volatility	0.1992

The objective of this second experiment is not to construct a tradable option pricing model or a production-grade calibration. Instead, it demonstrates that the same workflow can be applied using observed financial data rather than synthetic inputs.

Pricing results obtained from the three methods are reported in Table 3.5 and visualized in Figure 3.6.

Table 3.5: Option pricing results using market-derived inputs from AAPL historical data.

Method	Option Price	Absolute Error
Black-Scholes	19.8397	0.0000
Monte Carlo	19.6705	0.1693
Metropolis-Hastings	19.5573	0.2824

The overall conclusions are consistent with those obtained in the controlled experiment. The numerical estimates remain close to the Black-Scholes benchmark, while standard Monte Carlo continues to provide the most accurate approximation among the simulation-based methods considered. At the same time, the exercise demonstrates that the implementation can be applied without modification once the required market inputs have been estimated.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

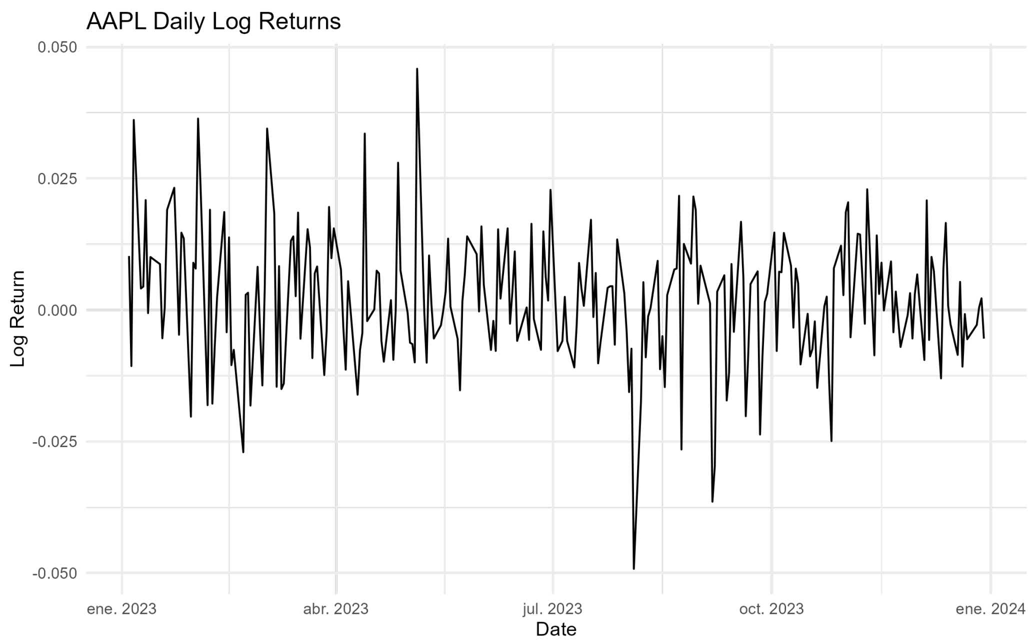

Figure 3.5: Historical log returns used to estimate the volatility parameter employed in the pricing exercise.

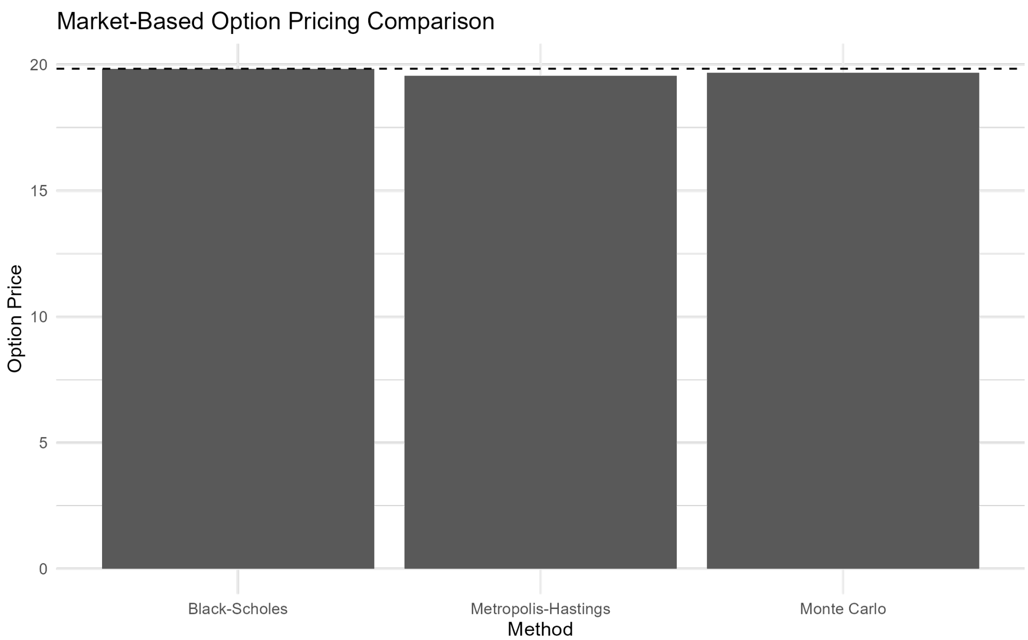

Figure 3.6: Comparison of pricing estimates obtained using market-derived inputs from AAPL historical data.

3.4 Results, Discussion, and Limitations

The results show that all three pricing approaches considered in this project produce estimates that are reasonably close to one another. In both the controlled experiment and the market-data example, the numerical estimators converge toward values consistent with the Black-Scholes benchmark, indicating that the implementation captures the essential characteristics of the pricing problem.

For the baseline European call option, standard Monte Carlo simulation produced the most accurate numerical approximation among the simulation-based methods. The Metropolis-Hastings estimator also generated reasonable price estimates, but exhibited larger deviations from the analytical solution and greater variability during the convergence analysis.

This outcome is not particularly surprising. The problem considered here is intentionally simple. Under the Black-Scholes assumptions, the terminal distribution of the asset price is known, making direct Monte Carlo simulation both straightforward and efficient. In this setting, introducing a Markov Chain Monte Carlo procedure adds computational complexity without providing a corresponding improvement in accuracy.

From a practical perspective, this observation is one of the most important findings of the project. The results suggest that Metropolis-Hastings should not be viewed as a replacement for standard Monte Carlo methods in simple European option pricing problems. When direct sampling is available, conventional Monte Carlo remains the more natural and efficient choice.

The convergence analysis reinforces this conclusion. As the number of samples increases, both estimators move toward the analytical benchmark. However, the Monte Carlo estimator exhibits a smoother convergence pattern, while the Metropolis-Hastings estimates fluctuate more noticeably at smaller sample sizes. This behavior is consistent with the autocorrelation inherent to Markov chain sampling and highlights one of the practical trade-offs associated with MCMC-based methods. Although Markov chains can explore complex probability distributions, successive samples are not independent, which can reduce sampling efficiency.

At the same time, the fact that Metropolis-Hastings does not outperform Monte Carlo in this experiment does not imply that MCMC methods are uninteresting for option pricing. The primary motivation for studying these techniques arises in situations where direct sampling becomes difficult or inefficient. Many derivatives encountered in practice depend not only on the terminal asset price but on the entire trajectory followed by the underlying asset. In such cases, the dimensionality of the problem increases substantially, and sampling-based approaches inspired by path integrals may become more attractive.

The market-data example provides an additional perspective on the methodology. Using historical AAPL prices, the same workflow produced pricing estimates that remained close to the analytical benchmark. While this exercise should not be interpreted as a production-grade option valuation framework, it demonstrates that the implementation can be applied directly to observed market data once the required inputs have been estimated. The transition from synthetic examples to real financial data required no modifications to the underlying pricing algorithms, which is an encouraging indication of the flexibility of the approach.

Several limitations should be acknowledged. First, the project focuses exclusively on European call options under the Black-Scholes assumptions. These assumptions include constant volatility, a constant risk-free interest rate, frictionless markets, and geometric Brownian motion dynamics. While these simplifications make the problem analytically tractable, they do not fully capture the behavior of real financial markets.

Second, the implementation does not perform a full path-integral simulation. The theoretical discussion motivating the project is expressed in terms of asset price trajectories, but the numerical implementation exploits the fact that European option payoffs depend only on the terminal asset price. As a result, Metropolis-Hastings is applied to the terminal log-price distribution rather than to complete asset paths. This simplification is appropriate for the problem studied here, but it leaves open the question of how the methodology would behave when applied to genuinely path-dependent derivatives.

Finally, the market-data example relies on historical volatility estimated from past returns. In practice, option pricing often incorporates additional sources of information such as implied volatility surfaces, stochastic volatility models, local volatility models, or alternative asset dynamics. Extending the framework to accommodate these features would require a more sophisticated calibration procedure than the one considered in this project.

Despite these limitations, the project successfully demonstrates that concepts originating from the path integral formulation can be translated into a practical computational workflow. For simple European options, the results confirm that standard Monte Carlo simulation remains the most efficient numerical approach among those considered. However, the implementation also provides a foundation for exploring more complex derivatives, where path-dependent payoffs and higher-dimensional probability distributions may create conditions under which MCMC-based techniques become more compelling.

3.5 Technical Appendix (Optional)

The main body of this project focuses on the practical implementation and evaluation of path-integral-inspired option pricing methods. The purpose of this appendix is to provide additional mathematical and computational details supporting the numerical results presented throughout the chapter.

The material is organized around four topics. Appendix A reviews the Black-Scholes framework underlying the pricing model. Appendix B introduces the Monte Carlo estimator used to approximate risk-neutral expectations. Appendix C summarizes the Metropolis-Hastings algorithm employed in the numerical experiments. Finally, Appendix D discusses the path integral perspective motivating the project and its connection to more general path-dependent pricing problems.

The appendix is intended as a technical reference rather than a prerequisite. Readers interested primarily in the implementation and results can safely skip this section without affecting the overall understanding of the project.

3.5.1 Appendix A. Black-Scholes Framework

The numerical experiments presented in this project are built upon the Black-Scholes framework developed by Black and Scholes and later extended by Merton [16] [17]. This appendix summarizes the mathematical assumptions underlying the model and connects them to the numerical implementation.

The starting point is the assumption that the underlying asset follows a geometric Brownian motion. The asset price dynamics are described by

$$dS_t = \mu S_t, dt + \sigma S_t, dW_t \tag{3.1}$$

where S_t denotes the asset price at time t, μ is the drift parameter, σ is the volatility, and W_t represents a standard Wiener process.

Equation 3.1 states that the evolution of the asset price contains both a deterministic component and a random component. The deterministic term controls the average growth of the process, while the stochastic term introduces uncertainty into future prices and generates the dispersion observed in the simulated trajectories presented in the main body of the project.

Applying Itô's lemma to the logarithm of the asset price yields a stochastic process with a closed-form solution. The terminal asset price can then be written as

$$S_T = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right) T + \sigma\sqrt{T}, Z\right] \tag{3.2}$$

where $Z \sim N(0,1)$ is a standard normal random variable.

Equation 3.2 implies that the logarithm of the terminal asset price is normally distributed,

$$\log(S_T) \sim N\left(\log(S_0) + \left(\mu - \frac{\sigma^2}{2}\right) T, \sigma^2 T\right) \tag{3.3}$$

which means that the terminal asset price itself follows a lognormal distribution.

This result plays a central role throughout the project. The Monte Carlo simulations generate terminal prices using Equation 3.2, while the Metropolis-Hastings implementation samples from the distribution described by Equation 3.3. As discussed in the main body of the project, the payoff of a European call option depends only on the terminal asset price, making this distribution the key object of interest.

Under the risk-neutral valuation framework, the expected growth rate of the asset is replaced by the risk-free interest rate r. The value of a European call option can then be expressed as

$$C = e^{-rT}\mathbb{E}\left[\max(S_T - K, 0)\right] \tag{3.4}$$

where C denotes the option value and K is the strike price.

Evaluating Equation 3.4 under the lognormal distribution of S_T leads to the Black-Scholes formula

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \tag{3.5}$$

with

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{3.6}$$

and

$$d_2 = d_1 - \sigma\sqrt{T} \tag{3.7}$$

Here, $N(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

For the purposes of this project, the availability of an analytical solution is particularly valuable. It provides a reference against which the Monte Carlo and Metropolis-Hastings estimators can be evaluated, allowing numerical errors and convergence behavior to be studied in a controlled setting. The Black-Scholes model therefore serves not only as a pricing framework, but also as a benchmark for validating the numerical methods explored throughout the chapter.

3.5.2 Appendix B. Monte Carlo Option Pricing

The Black-Scholes pricing formula provides a closed-form solution for European call options. However, many derivatives do not admit analytical solutions, making numerical methods necessary. One of the most widely used approaches in quantitative finance is Monte Carlo simulation [18].

The starting point is the risk-neutral valuation formula introduced in Equation 3.4,

$$C = e^{-rT}\mathbb{E}\left[\max(S_T - K, 0)\right] \tag{3.8}$$

This expression states that the option value is equal to the discounted expected payoff under the risk-neutral probability measure. In most practical applications, the expectation cannot be evaluated analytically and must be approximated numerically.

Monte Carlo simulation replaces the theoretical expectation with an empirical average computed from a large number of simulated outcomes. If $S_T^{(1)}, S_T^{(2)}, \dots, S_T^{(N)}$ denote N independent realizations of the terminal asset price, the option value can be approximated by

$$\hat{C}_N = e^{-rT} \frac{1}{N} \sum_{i=1}^{N} \max\left(S_T^{(i)} - K, 0\right). \tag{3.9}$$

Equation 3.9 is the estimator implemented in the Python notebook. Each simulation generates a possible terminal asset price according to the geometric Brownian motion model described in Appendix A, computes the corresponding payoff, and averages the results across all simulated scenarios.

An important property of Monte Carlo methods is that the estimator converges to the true expected value as the number of simulations increases. Under standard regularity conditions,

$$\hat{C}_N \to C \qquad \text{as} \qquad N \to \infty. \tag{3.10}$$

This result follows from the Law of Large Numbers and provides the theoretical justification for the simulation approach.

The accuracy of the estimator is commonly measured through its standard error. For Monte Carlo methods, the estimation error decreases proportionally to

$$\mathcal{O}\left(N^{-1/2}\right), \tag{3.11}$$

which implies that reducing the error by a factor of two requires approximately four times as many simulations.

This characteristic is one of the principal strengths and weaknesses of Monte Carlo methods. On the one hand, the approach is conceptually simple and can be applied to a wide range of pricing problems. On the other hand, convergence can be relatively slow, particularly when high precision is required.

For the European call option considered in this project, Monte Carlo simulation performs remarkably well. The results presented in the main body of the chapter show that the numerical estimates converge rapidly toward the Black-Scholes benchmark as the number of simulations increases. This behavior is expected because the terminal distribution of the underlying asset is known and can be sampled directly.

The role of Monte Carlo simulation in this project is therefore twofold. First, it provides a practical numerical approximation to the Black-Scholes pricing problem. Second, it serves as a baseline against which the Metropolis-Hastings implementation can be evaluated. Since direct sampling is available in this setting, Monte Carlo represents a natural benchmark for assessing the advantages and limitations of alternative sampling-based approaches.

3.5.3 Appendix C. Metropolis-Hastings Sampling

The Monte Carlo estimator discussed in Appendix B relies on direct sampling from the target distribution. In the Black-Scholes setting considered in this project, this approach is straightforward because the terminal asset price distribution is known analytically. However, many problems in computational finance involve probability distributions that are difficult to sample from directly.

Markov Chain Monte Carlo (MCMC) methods address this challenge by constructing a stochastic process whose long-run behavior reproduces a desired target distribution. Rather than generating independent samples, MCMC algorithms produce a sequence of correlated samples that gradually explore the probability space.

The Metropolis-Hastings algorithm is one of the most widely used MCMC methods. The algorithm begins from an initial state x and proposes a new candidate state x' from a proposal distribution $q(x'|x)$. The proposed state is then accepted with probability

$$\alpha(x, x') = \min\left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}\right), \tag{3.12}$$

where $\pi(\cdot)$ denotes the target probability distribution.

If the proposed state is accepted, the Markov chain moves to x'. Otherwise, the chain remains at its current position. Repeating this procedure produces a sequence of samples whose stationary distribution converges to the target distribution.

In the present project, the objective is not to sample complete asset-price trajectories. Instead, the algorithm is applied to the terminal log-price distribution derived in Equation 3.3. This simplification is possible because the payoff of a European call option depends only on the terminal asset price.

The target distribution therefore corresponds to

$$\log(S_T) \sim N\left(\log(S_0) + \left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right), \tag{3.13}$$

where the drift parameter has been replaced by the risk-free interest rate in accordance with the risk-neutral valuation framework.

Once samples from the terminal distribution have been generated, option values can be estimated using the same discounted payoff expression introduced in Appendix B. The difference lies not in the pricing equation itself, but in the mechanism used to obtain the samples.

An important consequence of this approach is that successive observations generated by the Markov chain are not independent. This property introduces autocorrelation into the sample and can reduce statistical efficiency relative to direct Monte Carlo simulation.

This observation helps explain the behavior reported in the main body of the project. The Metropolis-Hastings estimator produced option values consistent with the Black-Scholes benchmark, but exhibited larger fluctuations than standard Monte Carlo at smaller sample sizes. Since direct sampling from the terminal distribution is available in this problem, the additional flexibility of MCMC provides little practical advantage.

The purpose of including Metropolis-Hastings in this project is therefore not to outperform conventional Monte Carlo simulation. Rather, it serves as an introduction to sampling techniques that become increasingly relevant when direct sampling is difficult or when the pricing problem depends on higher-dimensional probability distributions. These situations arise naturally in many path-dependent derivatives and motivate the broader path integral perspective discussed in the next appendix.

3.5.4 Appendix D. Path Integral Perspective

The title of this project refers to the path integral formulation of option pricing. Although the numerical implementation developed throughout the chapter does not perform a full path integral simulation, the underlying motivation originates from a broader perspective on stochastic processes and probability distributions.

The path integral formalism was originally developed in quantum mechanics by Feynman as an alternative formulation of physical dynamics [19]. Rather than describing the evolution of a system through a single trajectory, the theory considers all possible trajectories connecting an initial and a final state. Observable quantities are then obtained by aggregating the contributions of these trajectories.

A similar idea appears naturally in stochastic finance. The future value of a derivative depends on the possible evolutions of the underlying asset, each associated with a particular probability. From this viewpoint, option pricing can be interpreted as an average over a space of possible asset-price paths [20].

In a schematic form, the value of a derivative can be expressed as

$$C(S_0, t_0) = \int \mathcal{D}[S(t)], P[S(t)], \Phi[S(t)] \tag{3.14}$$

where $\mathcal{D}[S(t)]$ denotes integration over all possible trajectories, $P[S(t)]$ represents the probability associated with a given path, and $\Phi[S(t)]$ denotes the payoff functional.

The key distinction between this expression and the formulations presented in the previous appendices is that the object being integrated is no longer a single terminal value, but an entire trajectory.

For the European call option considered in this project, the payoff depends only on the terminal asset price,

$$\Phi[S(t)] = \max(S_T - K, 0). \tag{3.15}$$

As a consequence, the pricing problem can be reduced to the terminal distribution of S_T. This observation explains why the implementation developed in the chapter focuses on sampling terminal prices rather than complete paths. For this particular derivative, the additional information contained in the intermediate trajectory is unnecessary for computing the payoff.

This simplification is one of the reasons why standard Monte Carlo simulation performs so well in the experiments presented earlier. Direct sampling from the terminal distribution is available, making more sophisticated sampling strategies difficult to justify on efficiency grounds alone.

The situation changes when the payoff depends on the entire history of the asset rather than its terminal value. Examples include Asian options, lookback options, barrier options, and many other path-dependent derivatives. In these cases, the trajectory itself becomes part of the pricing problem, and the path integral viewpoint provides a natural conceptual framework for describing the associated probability space [21] [22].

From this perspective, the implementation developed in this project can be viewed as a simplified introduction to a broader class of numerical techniques. The Monte Carlo and Metropolis-Hastings algorithms operate on the terminal distribution because the European call option permits this reduction. More complex derivatives would generally require sampling higher-dimensional objects, including entire asset-price trajectories.

The objective of this chapter was not to build a complete path integral pricing engine, but rather to investigate whether ideas originating from the path integral framework could be translated into a practical computational workflow. The results suggest that, for simple European options, direct Monte Carlo simulation remains the most efficient approach among those considered. At the same time, the project establishes a foundation for exploring more sophisticated derivatives in which the trajectory itself becomes the central object of interest.

Part II

Retail Analytics

Overview

This part covers problems related to retail analytics, including the analysis of customer behavior, transaction data, and marketing performance.

The scope of this domain includes methods for segmentation, prediction, and decision support, with applications in areas such as fraud detection, customer lifetime value estimation, and campaign optimization.

Chapter 4

Credit Card Fraud Detection

Every credit card transaction generates a simple decision: approve it or block it. Most of the time the answer is obvious. The difficulty lies in the small fraction of transactions that appear legitimate but are actually fraudulent. In large payment systems, these events are rare, costly, and constantly changing, creating a learning problem that differs substantially from many conventional classification tasks.

This project uses a publicly available European credit card transaction dataset containing highly anonymized transaction records. The analysis combines unsupervised learning techniques to explore the structure of fraudulent activity with a supervised benchmarking framework designed to evaluate the predictive performance of several classification algorithms.

4.1 Problem Definition

The central difficulty in credit card fraud detection is severe class imbalance. Fraudulent transactions represent only a tiny fraction of overall activity, meaning that a model can achieve high accuracy while failing to identify most fraud cases [23].

This imbalance creates an unavoidable trade-off. Missing a fraudulent transaction may result in direct financial losses, while incorrectly flagging a legitimate transaction may interrupt a customer purchase and generate unnecessary friction. Effective fraud detection therefore requires more than maximizing a single performance metric. The objective is to identify as many fraudulent transactions as possible while maintaining an acceptable level of false alarms [24] [25].

Most machine learning studies approach this problem through supervised classification, using historical examples of fraudulent and legitimate transactions to train predictive models. This approach has produced substantial improvements over traditional rule-based systems and remains the foundation of modern fraud analytics [26] [27].

A less explored question is whether fraudulent transactions exhibit recognizable structures before any classification model is trained. If fraud is not a single behavior but a collection of different behaviors, clustering techniques may reveal groups of transactions that share common characteristics. Such patterns could provide additional insight into how fraudulent activity is organized within the feature space and help explain why certain classification models perform better than others [28].

The project therefore investigates the problem from two complementary perspectives. First, clustering methods are used to explore whether fraudulent transactions form identifiable groups within the feature space. Second, a supervised benchmarking study compares several machine learning algorithms under a common evaluation framework. The objective is not only to determine which models achieve the strongest predictive performance, but also to better understand the structure of fraudulent behavior in a highly imbalanced transaction dataset.

4.2 Modeling Approach

The analysis combines unsupervised and supervised learning techniques to examine credit card fraud from two complementary perspectives. The first investigates whether fraudulent transactions exhibit identifiable structure within the feature space. The second evaluates the extent to which that structure can be exploited for predictive purposes.

The unsupervised stage applies clustering algorithms to the transaction data in order to explore the distribution of fraudulent and legitimate transactions. K-Means, Agglomerative Clustering, and DBSCAN are used because they represent different notions of similarity and cluster formation. K-Means partitions observations around centroids, Agglomerative Clustering builds nested groups through successive merges, and DBSCAN identifies groups based on local density while explicitly accounting for noise points. Together, these methods allow the analysis to examine whether fraudulent transactions form concentrated groups, distinct fraud profiles, or isolated anomalies.

The supervised stage is formulated as a benchmarking study. Rather than focusing on a single predictive model, the project compares a diverse set of classification algorithms representing different modeling philosophies. The benchmark includes Logistic Regression, Naive Bayes, Linear and Quadratic Discriminant Analysis, K-Nearest Neighbors, Linear Support Vector Machines, Decision Trees, Random Forests, and XGBoost. To examine the impact of class imbalance, the models are evaluated under three training strategies: the original transaction distribution, random undersampling, and the Synthetic Minority Oversampling Technique (SMOTE).

Because fraudulent transactions represent only a small fraction of the observations, model evaluation focuses on metrics that remain informative under severe class imbalance. Particular attention is given to precision, recall, F1-score, ROC-AUC, Precision-Recall AUC, and the Kolmogorov-Smirnov statistic. Stratified cross-validation is used throughout the benchmark to ensure consistent comparisons across models.

The underlying assumption of the project is that fraudulent transactions leave detectable signatures in the available features. If such signatures exist, clustering methods should reveal some degree of structure within the transaction space, while supervised learning algorithms should be able to exploit that structure for predictive purposes. Examining both perspectives provides a more complete view of fraud behavior than either approach in isolation.

4.3 Methodology and Implementation

The analysis begins with an unsupervised exploration of the feature space to determine whether fraudulent transactions exhibit identifiable structure without using the target labels. Clustering algorithms are used to examine the distribution of transactions and to assess whether fraud tends to concentrate in specific regions of the data.

The second stage introduces the transaction labels and evaluates a range of supervised learning algorithms within a common benchmarking framework. Together, these two perspectives provide a broader understanding of the problem, combining exploratory analysis with predictive modeling.

4.3.1 Unsupervised Learning Methodology

The unsupervised learning stage investigates whether fraudulent transactions exhibit identifiable structure in the feature space without using the target labels during model construction. If fraudulent transactions tend to concentrate in specific regions of the transformed feature space, clustering algorithms should be able to reveal these patterns.

The analysis begins with an exploratory examination of the dataset. Particular attention is given to the severe class imbalance that characterizes the fraud detection problem.

The class distribution is summarized below.

Table 4.1: Class distribution in the credit card transaction dataset.

Class	Transactions	Percentage (%)
Legitimate	284,315	99.827
Fraudulent	492	0.173

Figure 4.1 illustrates the extreme imbalance between legitimate and fraudulent transactions. Fraudulent transactions represent less than one percent of all observations, making fraud detection a highly imbalanced classification problem.

Transaction amounts were also examined to characterize the range and concentration of transaction values.

Table 4.2: Summary statistics for transaction amounts.

Statistic	Value
Minimum	0.00
1st Quartile	5.60
Median	22.00
Mean	88.35
3rd Quartile	77.17
Maximum	25,691.16

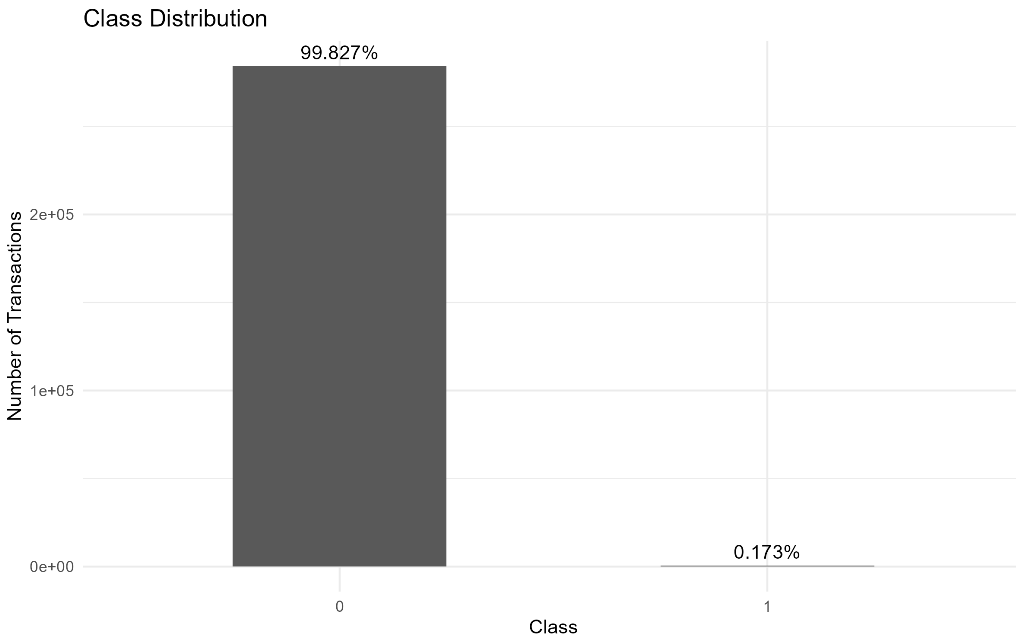

Figure 4.1: Class distribution in the credit card transaction dataset.

The distribution of transaction amounts is highly right-skewed, with most transactions concentrated at relatively small values and a small number of extreme observations. This behavior can be observed in Figure 4.2.

Transaction amounts were further compared across classes. Although fraudulent and legitimate transactions overlap substantially, Figure 4.3 suggests differences in the distribution of transaction values between the two groups.

The variables V1–V28 correspond to principal components obtained through a prior anonymization process. Selected features were visualized to examine differences between fraudulent and legitimate transactions. Several variables exhibit noticeable shifts between classes, suggesting that fraudulent transactions occupy distinct regions of the transformed feature space.

To complement the univariate analysis, a correlation matrix was computed using all available variables.

Following the exploratory analysis, the target variable was separated from the predictors and the Time variable was removed. The data was then partitioned into training and test sets using stratified sampling to preserve the original class

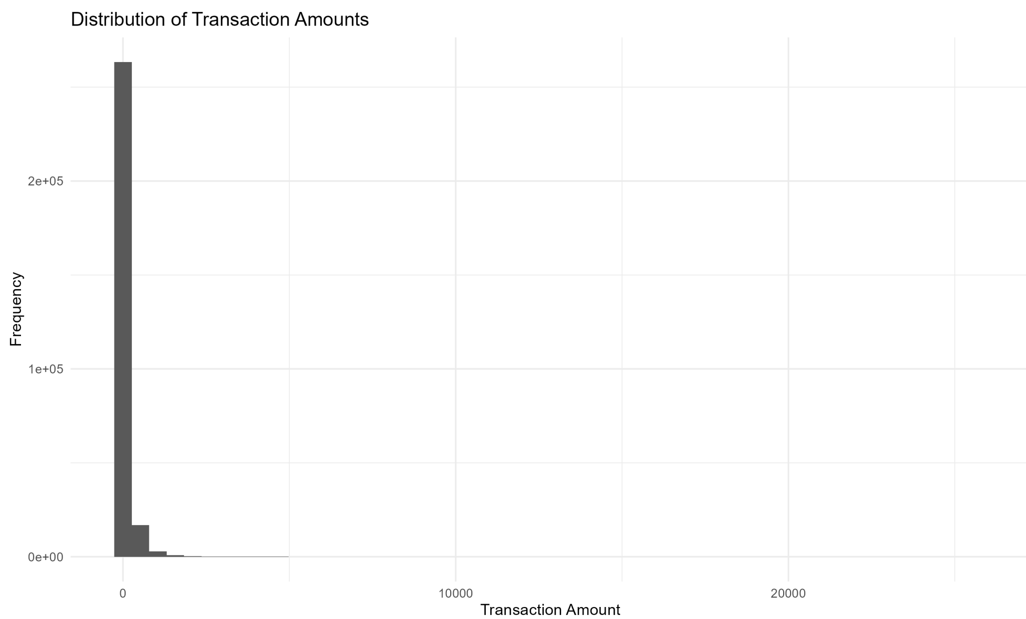

Figure 4.2: Distribution of transaction amounts.

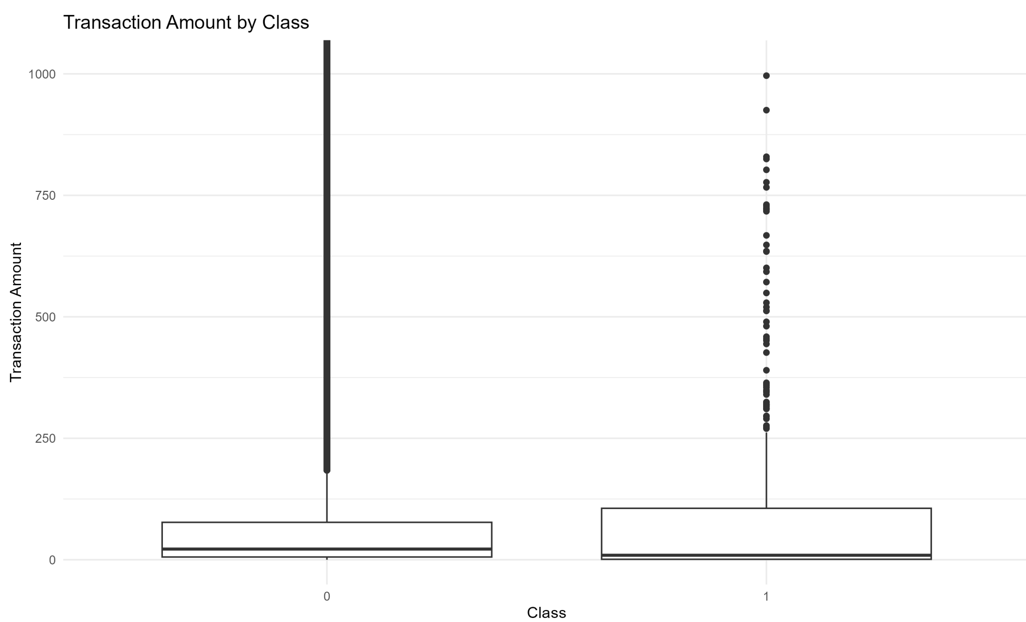

Figure 4.3: Transaction amount by class.

Figure 4.4: Selected PCA feature distributions by class.

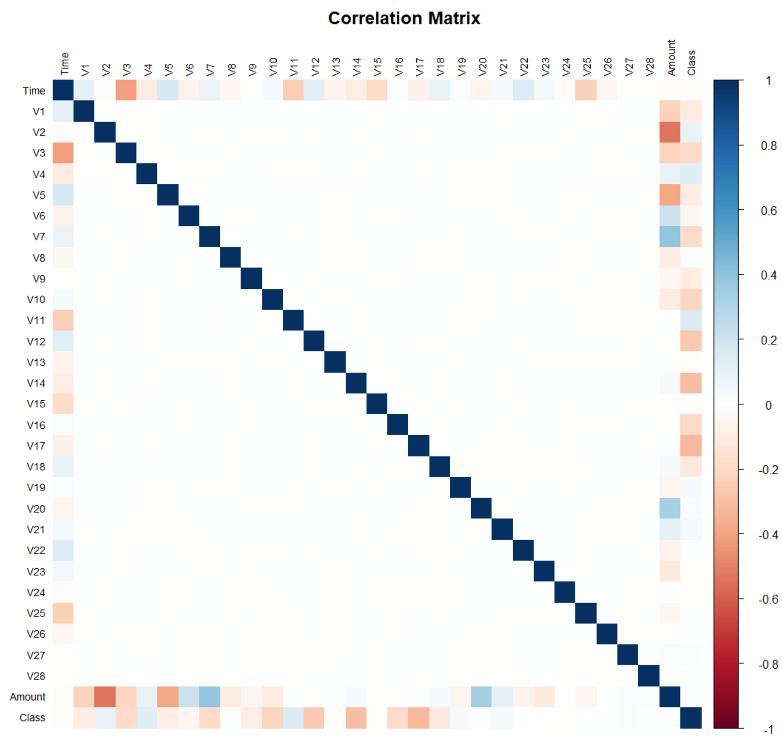

Figure 4.5: Correlation matrix of the credit card transaction dataset.

distribution. Feature scaling was applied after the train-test split, with scaling parameters estimated exclusively from the training data.

Because the training set contains more than two hundred thousand observations, a stratified sample was extracted from the training data for the clustering analysis. The sample preserves the original fraud rate while reducing computational requirements and allowing a consistent comparison across clustering algorithms.

Three clustering approaches were evaluated:

- K-Means Clustering
- Agglomerative Clustering
- DBSCAN

For K-Means, multiple values of (k) were explored. The elbow method was evaluated using the within-cluster sum of squares (inertia).

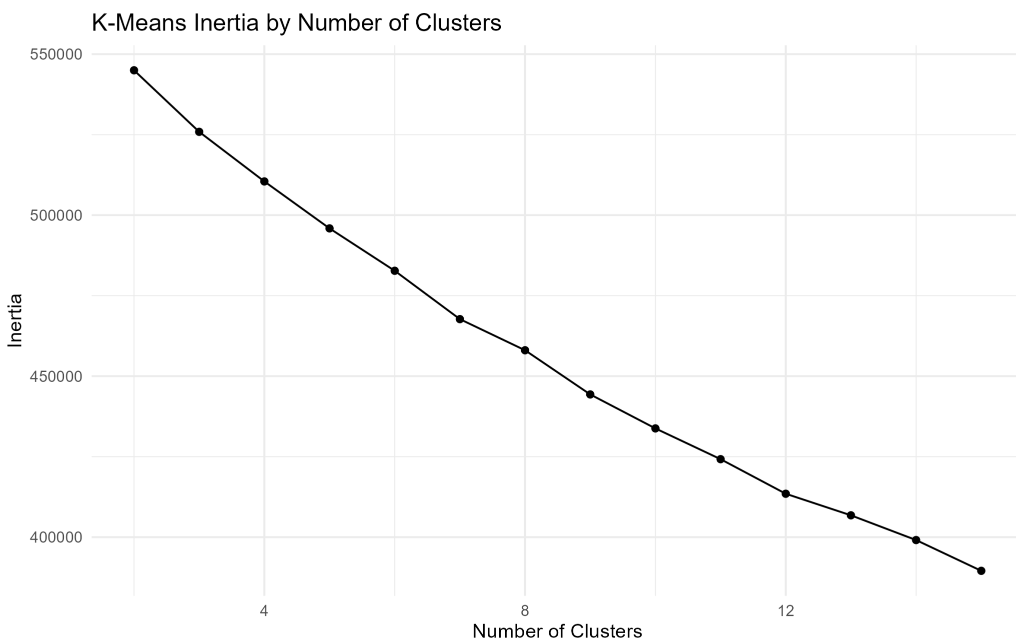

Figure 4.6: K-Means inertia as a function of the number of clusters.

The elbow plot does not reveal a clear point at which additional clusters provide diminishing returns. As a result, the number of clusters cannot be determined reliably using inertia alone. Additional validation metrics were evaluated and used as complementary diagnostic tools. Since no single criterion produced a definitive solution, the final choice was guided by interpretability and fraud concentration within the resulting clusters.

A three-cluster solution was retained because it revealed the strongest concentration of fraudulent transactions while remaining straightforward to interpret.

The composition of the resulting clusters was then examined by comparing cluster-level fraud rates against the baseline fraud rate observed in the clustering sample.

Table 4.3: Summary of the main findings obtained from the clustering algorithms.

Method	Key Finding
K-Means	One cluster exhibited a fraud rate approximately 21 times higher than the baseline fraud rate.
Agglomerative Clustering	A small cluster exhibited a fraud rate approximately 490 times higher than the baseline fraud rate.
DBSCAN	Fraudulent transactions were primarily identified as noise points rather than members of dense clusters.

The clustering results provide complementary perspectives on the structure of fraudulent transactions. K-Means identified a relatively large region characterized by elevated fraud concentration, while Agglomerative Clustering isolated a much smaller cluster with an exceptionally high fraud rate. In contrast, DBSCAN classified fraudulent transactions primarily as noise, suggesting that some fraud patterns behave as isolated anomalies rather than members of dense groups.

Taken together, these results indicate that fraudulent transactions are not uniformly distributed across the feature space. Although the exact structure varies across clustering algorithms, all three methods reveal behavior that differs substantially from the baseline distribution of legitimate transactions.

These findings provide preliminary evidence that fraudulent transactions leave detectable signatures in the available feature space and motivate the supervised learning stage of the analysis.

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

4.4 Supervised Learning Benchmark

Additional implementation details, including the companion Quarto and Python notebooks, are available in the project repository

4.5 Results, Discussion, and Limitations

4.6 Technical Appendix (Optional)

Chapter 5

Customer Value and Retention

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5.2 Modeling Approach

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5.3 Methodology and Implementation

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5.5 Technical Appendix (Optional)

Section.

Chapter 6

Customer Sentiment and Topic Analysis

Header.

6.1 Problem Definition

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6.2 Modeling Approach

Section.

6.3 Methodology and Implementation

Section.

6.4 Results, Discussion, and Limitations

Section.

6.5 Technical Appendix (Optional)

Section.

Part III

Biological Systems

Overview

This part covers problems related to biological systems, including the analysis and modeling of data arising in biological, health, and life science contexts.

The scope of this domain includes approaches for studying complex processes, identifying patterns in biological data, and supporting analysis in settings where systems are often heterogeneous, nonlinear, and influenced by multiple interacting factors.

Chapter 7

Predicting Crop Performance from Soil Microbial Communities

Header.

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Chapter 8

Medical Image Analysis for Skin Lesion Triage

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8.2 Modeling Approach

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8.3 Methodology and Implementation

Section.

8.4 Results, Discussion, and Limitations

Section.

8.5 Technical Appendix (Optional)

Section.

Part IV

Energy and Industrial Systems

Overview

This part covers problems related to energy and industrial systems, including the analysis and modeling of production processes, physical assets, and operational data.

The scope of this domain includes approaches for optimization and efficient resource allocation, monitoring and analysis of system integrity, and the study of degradation processes such as corrosion, as well as data-driven methods to support decision-making in complex operational environments.

Chapter 9

Permeability Prediction in Porous Media

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Chapter 10

Predicting Pipeline Corrosion for Asset Integrity Management

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10.5 Technical Appendix (Optional)

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Part V

Climate and Environmental Systems

Overview

This part covers problems related to climate and environmental systems, including the analysis and modeling of temporal and spatial data describing atmospheric and large-scale natural processes.

The scope of this domain includes approaches for time series analysis, forecasting, and the study of variability and uncertainty in systems influenced by natural dynamics, external drivers, and interactions across multiple scales.

Chapter 11

Sediment Source Modeling

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Chapter 12

Forecasting Drought Risk from Climate Time Series

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12.2 Modeling Approach

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Section.

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Section.

12.5 Technical Appendix (Optional)

Section.

Part VI

Writings & Essays

Overview

This part covers technical writings and essays spanning scientific modeling, data analysis, and related areas of mathematics, physics, and philosophy.

The scope includes discussions of modeling principles, reasoning under uncertainty, and computational approaches, providing context and deeper insight into the ideas underlying the projects.

Chapter 13

Data Science in the Age of Hype: Why the scientific method still matters

13.1 The Rise of Data Science

Data has become one of the most valuable assets of any modern organization. The unprecedented flow of information—resulting from countless digital interactions such as clicks, e-commerce transactions, mobile communications, and social media activity—reflects its central role in contemporary society. At the same time, the rapid advances in computing that enable large-scale data collection, storage, and processing are radically transforming the way businesses, governments, scientific communities, and institutions make decisions.

Yet, raw data by itself is meaningless. It only acquires value when subjected to systematic and rigorous analysis through methodologies capable of detecting patterns, inferring relationships, predicting trends, and ultimately translating insights into decisions with real-world impact. This is why Clive Humby's now-famous analogy describes data as "the new oil": like crude oil, data must be refined before it can generate value. The crucial element is not the "raw material" itself but the techniques used to extract its potential and the knowledge derived from it. This reflection naturally leads to a fundamental question: what do we really mean by Data Science?

Historically, the term Data Science emerged from an evolutionary process shaped by multiple contributions that helped formalize it as a discipline. Although its roots lie in statistics and computer science, its consolidation as an autonomous field took place over more than half a century of theoretical and technological developments.

The conceptual seed of Data Science can be traced back to 1962, when American statistician John W. Tukey published his influential essay *The Future of Data Analysis*. In it, Tukey argued that statistics should not be confined to hypothesis confirmation alone, but should place equal emphasis on the exploration, diagnosis, and understanding of data [29], [30]. He introduced *data analysis* as a distinct discipline, anticipating the vision of a "science of learning from data"—one in which empirical inquiry and iterative interaction with data occupy the core of quantitative knowledge.

Later, in 1974, Danish computer scientist Peter Naur, in his *Concise Survey of Computer Methods*, explicitly used the terms *Data Science* and *Datalogy* to refer to the systematic study of data [31]. Naur argued that computing should focus on the structure, meaning, and organization of data rather than on hardware or programming. Although his proposal went largely unnoticed for years, it laid conceptual foundations that would be revisited decades later.

In 1997, statistician C.F. Jeff Wu, during his inaugural lecture at the University of Michigan titled *Statistics = Data Science?*, proposed renaming statistics as *Data Science* and calling modern statisticians *data scientists* [32]. His goal was to signal a new era in which statistical inference should integrate with computation and empirical analysis to move beyond mere data description and outdated stereotypes.

This perspective was further developed in 2001 by William S. Cleveland in his seminal article *Data Science: an Action Plan for Expanding the Technical Areas of the Field of Statistics* [33]. Cleveland proposed a systematic expansion of statistics by identifying six core technical areas of Data Science: multidisciplinary investigations, models and methods for data, computing with data, pedagogy, tool evaluation, and theory. Rather than allowing statistics to contract into a narrowly mathematical discipline, he emphasized the central role of computation, real-world data analysis, education, and the development and evaluation of tools. His framework laid the conceptual groundwork for understanding Data Science as a comprehensive, data-centered field—deeply rooted in statistics, yet necessarily broader in scope and practice.

That same year, Leo Breiman, a statistician at the University of California, Berkeley, published his influential essay *Statistical Modeling: The Two Cultures*, in which he argued that statistical modeling serves two fundamental purposes: **information**—extracting insight into how nature associates response variables with input variables—and **prediction**—the ability to accurately anticipate future responses for new observations [34]. Breiman critically observed that traditional statistical practice had largely privileged the informational goal through highly structured, assumption-driven models, often at the expense of predictive performance. He advocated for a broader methodological perspective in which algorithmic and data-driven approaches play a central role, emphasizing prediction as an essential complement rather than a secondary objective. This distinction, and the tension it revealed between explanation and prediction, foreshadowed core principles of modern Data Science, where understanding and predictive accuracy are treated as intertwined but distinct aims.

By the late 2000s, the rise of Big Data and the growing need for professionals capable of combining statistics, programming, and domain expertise fueled the popularization of both the term Data Science and the role of the data scientist. Media consolidation came in 2012, when Thomas H. Davenport and DJ Patil published *Data Scientist: The Sexiest Job of the 21st Century* in the *Harvard Business Review* [35]. Since then, Data Science has become a strategic discipline across both academia and industry.

Today, Data Science is recognized as an interdisciplinary field that integrates statistical, computational, and conceptual methods to extract knowledge, infer patterns, and generate predictions from data. In many ways, it represents the natural evolution of statistics a discipline now equipped to address complex phenomena in fields as diverse as economics, biology, marketing, and physics.

However, despite its relevance and expansion, the term *Data Science* has become somewhat diffuse, often distorted by media overexposure. This ambiguity calls for a deeper examination of its epistemological foundations: if we speak of a **Science** of Data, we must ask what makes a discipline a science?

From this perspective, Data Science can indeed be considered a science. Its scientific character lies in the systematic application of the scientific method to the study of data. The data scientist formulates hypotheses, analyzes observations, and tests results just as any other scientist does. The difference lies in the object of study: data as a quantitative representation of reality.

In this sense, Data Science can be viewed as a generalization of statistics extending its focus from **inference** (understanding how the world behaves based on observed data) to **prediction** (anticipating how it might behave under new conditions). This predictive capability, powered by machine learning algorithms, enables the effective application of the **Hypothetical-Deductive Method**: formulating hypotheses about future phenomena and testing them through data and predictive models. In other words, machine learning acts as a technological enabler that materializes Breiman's inference–prediction duality, providing tools to model complex relationships directly from data and to test hypotheses that were previously unreachable with traditional statistical methods. It is this integration of inference, prediction, and empirical validation that solidifies the scientific nature of the discipline.

Understanding these foundations is essential for interpreting the true scope of Data Science. While the discipline draws from statistics, computer science, and domain expertise, its ultimate objective is not merely the manipulation of data but the extraction of reliable knowledge about the world. In this sense, Data Science should not be understood as a collection of algorithms or software tools, but as a methodological framework for reasoning under uncertainty.

However, the rapid growth of the field has also brought an unintended consequence: a proliferation of narratives that portray Data Science primarily as a technological revolution driven by ever more sophisticated algorithms. In many contexts, the emphasis has shifted toward tools, automation, and predictive performance, sometimes overshadowing the epistemological principles that historically guided scientific inquiry.

This tension raises a critical question for modern practitioners: if Data Science claims to be a science, what role does the scientific method play in its practice today? Addressing this question requires revisiting the principles that define scientific reasoning and examining how they apply to the analysis and interpretation of data in an era increasingly shaped by computational power and algorithmic models.

13.2 The Nature of Science and the Scientific Method

The very semantics of the term *Data Science* make it impossible to detach it from the concept of *science*. This forces us to pause and reflect: can Data Science truly be regarded as a scientific discipline, or is it merely a fashionable commercial label? A comprehensive analysis of the nature of science lies far beyond the scope of this article; however, for the sake of intellectual honesty, we should not evade the issue. At the very least, we ought to clarify what makes an activity genuinely *scientific*.

From an epistemological perspective, science cannot be defined merely as any endeavor that leads to new discoveries or expands knowledge. Scientific knowledge evolves what was once considered true has often been revised, refined, or discarded. It is therefore reasonable to assume that part of what we regard today as scientifically valid will eventually be replaced, even if we cannot foresee which part. In other words, science does not guarantee truth. What distinguishes science from other forms of inquiry is not the *results* it produces, but the *method* it employs to generate them [36]. Science is better understood as an iterative mechanism—almost algorithmic in nature—that produces, tests, refines, and occasionally rejects knowledge. This is a central idea, because in essence, a data scientist does exactly that: applies the scientific method as a repeatable process of inquiry. We will return to this point shortly when discussing the *Hypothetico-Deductive Method*.

With this foundation in place, it becomes easier to see why Data Science can be viewed as a generalization of Statistics. This is not a trivial claim. Statistics is undoubtedly a science specifically, a formal science, alongside logic and computer science. Formal sciences study abstract systems through reasoning rather than direct experimentation, distinguishing them from natural, social, and applied sciences. Yet classical statistics has been historically grounded in *inductive inference*: reasoning from the particular to the general. Induction allows us to infer broader conclusions from observed cases, but it carries a fundamental limitation: inductive reasoning can *support* a conclusion, but never *prove* it. This problem, famously articulated by David Hume in the 18th century and known as the *problem of induction*, arises from the logical gap between past observations and future expectations.

Bertrand Russell illustrated this issue through the well-known "inductivist turkey". A turkey observes that every morning at 9 a.m. the farmer brings food. After hundreds of consistent observations—rainy days, sunny days, warm and cold alike—the turkey confidently concludes that it will always be fed at 9 a.m. Yet on Christmas Eve, instead of being fed, it is slaughtered. A single contradictory event collapses a conclusion seemingly supported by overwhelming evidence. The lesson is clear: no amount of past observations can logically ensure the truth of a universal claim about the future. Karl Popper pushed this idea further: induction can never confirm a theory with certainty, whereas a single counterexample can refute it. The turkey's mistake—rooted in the "turkey illusion"—was assuming regularity without understanding the underlying mechanism (the farmer's intention).

None of this implies that induction is a "bad" method. It remains essential as the generator of hypotheses and conjectures. As Klimovsky notes [37], induction may be seen as a probabilistic justification: it evaluates how strongly observations support a hypothesis, rather than certifying its truth. Put simply, science advances by combining induction (to propose hypotheses) and deduction (to test them). Deduction enables us to assess whether predictions derived from hypotheses correspond to reality. Verification, however, does not grant absolute truth; it merely confirms that a hypothesis has survived a test under certain conditions.

This tension lies at the heart of Leo Breiman's critique of classical statistics. While statistics provided the mathematical language to quantify uncertainty and draw inferences about populations from limited data, it remained fundamentally bound to inductive reasoning: no matter how large the sample or how robust the model, a future observation could always contradict it. Machine Learning transformed this landscape by not only inferring patterns from data but also validating predictive performance empirically. A model is no longer judged solely by how well it explains past data, but by how accurately it predicts *unseen* data. This represents a qualitative shift that operationalizes the hypothetico-deductive paradigm: generating hypotheses (models) and testing them empirically against new observations in a reproducible and quantifiable way.

To illustrate this in a Data Science context, consider a credit-scoring model. Classical statistics may estimate relationships between customer features and default risk through inductive inference on historical samples. A machine learning model, however, must *demonstrate its predictive power* on new customers those the model has never encountered. If the predictions fail, the model is rejected or refined. This is falsification in action, applied to data-driven decision-making.

In this sense, Data Science can be expressed as **Inference + Prediction**, or equivalently, as **Statistics + Machine Learning**. It unites the inductive strength of statistics with the deductive rigor of model testing, fulfilling Breiman's vision of a discipline that bridges explanation and prediction a modern science of learning from data. The next section will examine this duality through the lens of the *Hypothetico-Deductive Method*, the framework that ultimately solidifies the scientific character of Data Science.

13.3 The Scientific Mindset of a Data Scientist

At the core of scientific inquiry lies a logical structure that transcends individual disciplines: the **Hypothetico-Deductive Method**. It provides the conceptual backbone of the scientific method, linking theoretical conjecture with empirical observation. The process begins with a hypothesis—a tentative model proposed to explain observed behavior or predict its future course. or predict its future course. From this hypothesis, one deduces empirically testable consequences. These predictions are then confronted with real data through experimentation or systematic observation. When evidence contradicts the hypothesis, it must be revised or discarded; when the evidence supports it, the hypothesis gains provisional credibility, though never definitive truth.

This dynamic interplay between theoretical formulation, deduction, and empirical scrutiny defines the essence of scientific work. What grants scientific legitimacy to a model is not its mathematical elegance or computational sophistication, but its **capacity to generate falsifiable predictions**. The scientific method is therefore inherently self-correcting: each cycle of hypothesizing, testing, and refining moves knowledge closer to truth while acknowledging that truth remains an ever-distant horizon.

Traditional inferential statistics implements this logic only partially, primarily through hypothesis testing. A null hypothesis is proposed, its probabilistic implications are derived, and sample data are used to evaluate whether they justify its rejection. Yet this remains predominantly an inductive process, generalizing from finite observations into probabilistic claims about broader populations. Moreover, statistical validation is typically static once the data are gathered, the hypothesis is evaluated solely in that context, rather than through iterative confrontation with new evidence.

Machine learning, by contrast, operationalizes the deductive stage of scientific reasoning through **predictive validation**. Each model can be viewed as a hypothesis about the functional relationship between predictors and outcomes. Training a model corresponds to formulating the hypothesis; evaluating it on unseen data corresponds to testing its empirical consequences. Performance metrics quantify how well the model withstands empirical scrutiny, and failures become informative, guiding the reformulation or replacement of the underlying hypothesis.

This perspective turns Data Science into a **practical enactment of the Hypothetico-Deductive Method**. It combines inductive discovery of structure within data with deductive testing of the predictive consequences of that structure. More importantly, it enables Data Science to progress from mere correlation to **genuine explanation**, seeking models that capture mechanisms rather than patterns alone.

Consider the problem of **customer churn** in retail. A predictive model might accurately identify customers at risk of leaving based solely on behavioral correlations a purely inductive exercise. But if the goal is to determine whether a discount campaign will *reduce* churn, prediction alone is insufficient. The data scientist must investigate—and, when possible, formalize—**causal models** capturing how interventions alter outcomes. Prediction answers *who* is likely to churn; causality answers *why* and *what we can do about it*.

Under the Hypothetico-Deductive paradigm, this iterative cycle can be summarized visually, as shown in Figure 13.1, and is typically composed of the following stages:

- **Systematic observation** of a phenomenon
- **Formulation of hypotheses** or models to explain observed patterns
- **Deduction of testable predictions** grounded in those hypotheses
- **Empirical evaluation** of predictions via new data
- **Revision or replacement** of hypotheses based on the evidence collected

Viewed from this angle, the role of a data scientist extends far beyond executing algorithms or statistical routines. It is the **iterative application of the scientific method** across diverse business contexts generating hypotheses, building predictive and causal models, confronting them with evidence, and allowing data to arbitrate which explanations survive. Data Science thus emerges not merely as a technical discipline, but as the **contemporary expression of scientific thinking** applied to data-rich environments, where distinguishing correlation from causation and prediction from explanation defines true professional excellence.

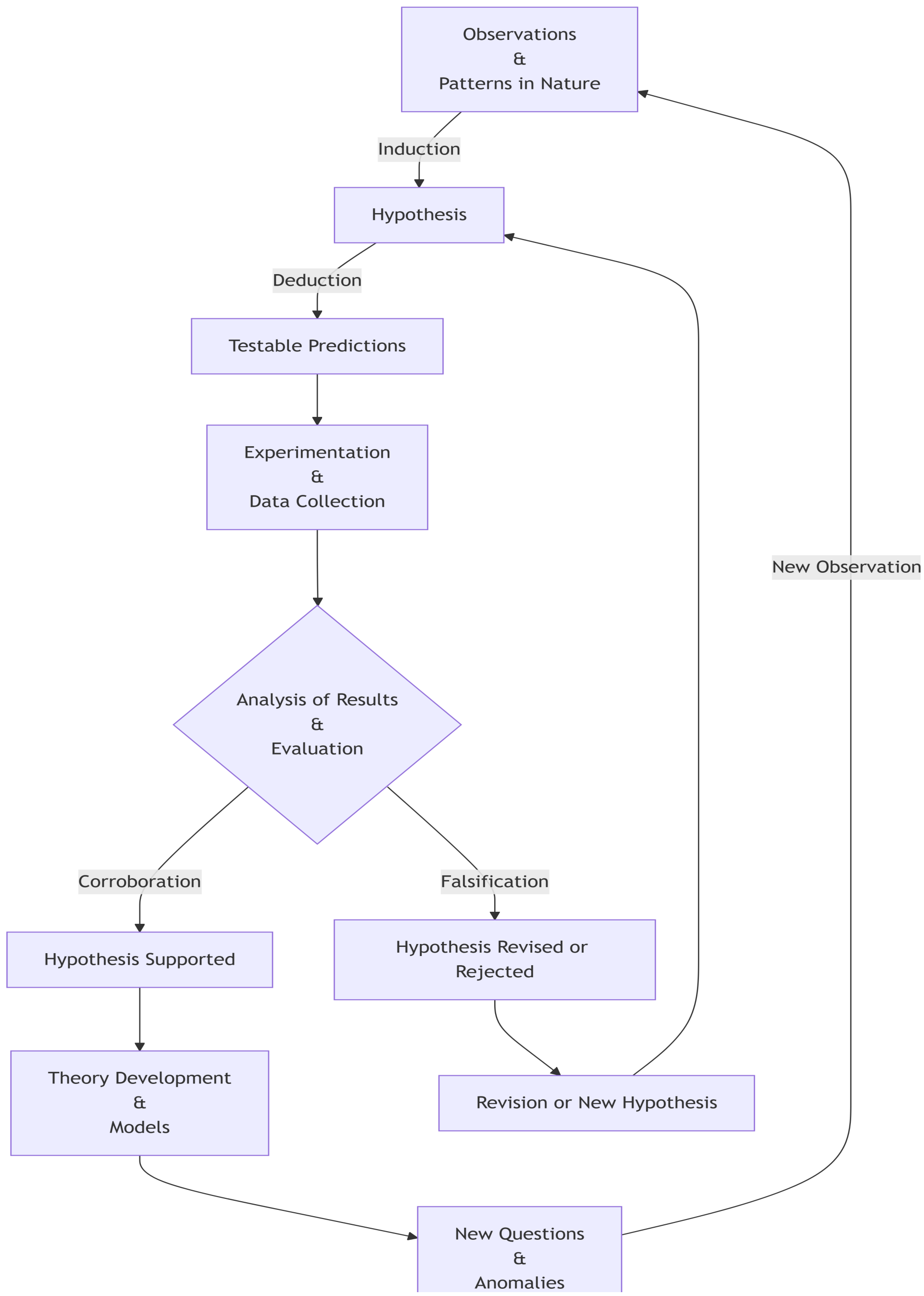

Figure 13.1: The hypothetico-deductive method as an iterative loop: observations motivate hypotheses (induction); hypotheses yield testable predictions (deduction); empirical results either support or challenge the hypothesis, leading to theory formation or revision and ultimately to new questions that restart the process.

13.4 The Fundamental Constituents of Data Science

Despite its rapid expansion and increasing visibility, Data Science is often portrayed as a discipline defined by an ever-growing collection of tools, frameworks, and technological trends. New libraries, platforms, and modeling techniques appear constantly, creating the impression that the field evolves primarily through technological novelty. This perception, however, can be misleading. While modern practice indeed relies on a rich computational ecosystem, the scientific core of Data Science is not defined by software stacks or fashionable methodologies. Rather, it rests on a far more compact and principled foundation: **probabilistic reasoning and statistical inference**.

From a first-principles perspective, Data Science is best understood as a systematic framework for reasoning under uncertainty using data. Probability provides the language to model randomness and variability, while Statistics supplies the methodological machinery for learning from data and evaluating the reliability of the conclusions we draw from it. Together, they form the conceptual backbone that allows data-driven reasoning to remain coherent, interpretable, and scientifically grounded.

Computation plays a crucial—but often misunderstood—role in this framework. The term *computation* derives from the Latin *computatio*, formed by the prefix *com* ("together" or "with") and *putatio*, related to *putare*, meaning "to calculate" or "to reckon." In its original sense, the word conveys the idea of **calculating together**. In this light, computation can be understood not as an autonomous source of knowledge, but as a process through which machines assist human reasoning by carrying out calculations at scales and speeds beyond our natural capacity. Algorithms, programming languages, and computational environments therefore act as instruments that enable probabilistic models to be simulated, statistical procedures to be implemented, and empirical hypotheses to be tested. Computation does not define Data Science; it provides the **operational medium** through which probabilistic and statistical reasoning becomes executable in practice.

13.5 Data Science as a Strategic Value Engine

At this point, it is crucial to emphasize a key principle that often gets overlooked: **Data Science is not a goal in itself**. No matter how sophisticated the methodologies or how advanced the machine learning models, they are worthless if they do not generate real and measurable impact. The true purpose of Data Science is to inform and improve strategic decisions—transforming data into outcomes that advance organizational goals.

Data-driven decision making empowers organizations to optimize operations, personalize customer experiences, identify risks proactively, and maintain a competitive advantage. As Kozyrkov highlights, outstanding data analysts ensure that their work is focused on solving the **right business problems**, not simply producing technically impressive outputs [38].

This underscores a critical truth: technical excellence without business alignment leads to solutions that may appear successful internally but **fail to influence decisions or create value**. The impact of a model is not measured solely by statistical accuracy or computational ingenuity, but by the extent to which it **changes what the organization does**.

From a decision-theoretic perspective, integrating Data Science into strategy enables measurable improvements not only in profitability or utility, but also in **decision velocity**. Automated analytical pipelines allow organizations to deploy models directly into operational environments continuously monitoring performance, adapting to new data, and discarding approaches that no longer deliver value. This creates a self-correcting loop where models are evaluated and refined over time, accelerating the pathway from insight to action.

Consider, for example, the case of **customer churn** in retail. Without Data Science, retention strategies typically rely on intuition or broad marketing campaigns, often costly and ineffective. In contrast, a churn prediction model can **identify customers most at risk**, enabling targeted interventions such as personalized offers or loyalty programs. If the strategy does not reduce churn among the targeted segment, performance monitoring mechanisms reveal the failure, prompting refinement or alternative plans. Here, data does not merely describe behavior—it actively shapes business outcomes.

Viewed through this lens, Data Science becomes a **strategic value engine**: a discipline that leverages scientific reasoning, computational tools, and domain expertise to drive decisions that matter. This synthesis—model-driven insight aligned with real-world objectives—ensures that Data Science fulfills its true mission: transforming information into impact.

13.6 Why the Scientific Method Still Matters

The strategic impact of Data Science ultimately stems from something deeper than predictive models or analytical pipelines. Its real strength lies in the intellectual framework that guides how data is interpreted, hypotheses are formulated, and decisions are evaluated. The ability of Data Science to generate meaningful impact in organizations is therefore inseparable from the scientific principles that structure its practice.

In an era defined by rapid technological change and the constant emergence of new tools, it is tempting to define Data Science by its most visible artifacts: programming languages, machine learning libraries, cloud platforms, and ever-evolving frameworks. Yet these elements, while powerful, are ultimately transient. Technologies evolve, tools are replaced, and methodologies shift as the field continues to mature.

What endures is something far more fundamental: the scientific mindset that underlies the discipline. At its core, Data Science is not about algorithms but about **reasoning under uncertainty**, formulating hypotheses, confronting them with data, and refining our understanding of complex phenomena through systematic evidence. Probability provides the language for uncertainty, statistics supplies the methods for inference, and computation serves as the medium through which these ideas become operational.

When practiced in this way, Data Science becomes more than a collection of analytical techniques—it becomes a modern expression of the scientific method applied to data-rich environments. Its true power lies not in automation or predictive accuracy alone, but in its capacity to transform data into knowledge and knowledge into better decisions. In the end, the enduring value of Data Science does not come from the sophistication of its tools, but from the rigor of the thinking that guides their use.

The original version is available on Medium here.

Chapter 14

Conformal Geometry and Electrostatics in Three Dimensions

Header.

14.1 Problem Definition

Section.

14.2 Modeling Approach

Section.

14.3 Methodology and Implementation

Section.

14.4 Results, Discussion, and Limitations

Section.

14.5 Technical Appendix (Optional)

Section.

14.6 Laplace's equation

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0 \tag{14.1}$$

14.7 Conformal Mappings

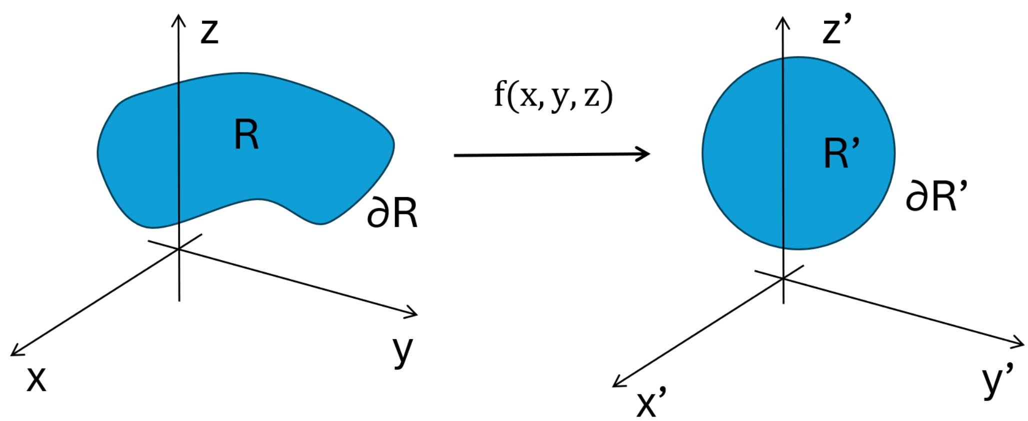

Figure 14.1: Schematic representation of a conformal transformation $f(x, y, z)$ mapping a region R with boundary ∂R in the (x, y, z) coordinate system into a transformed region R' with boundary $\partial R'$ in the (x', y', z') coordinate system.

As shown in Figure 14.1, ...

Suppose we have a region R in xyz space in which a scalar field ϕ satisfies Laplace's equation Equation 14.1. We want to determine what conditions a transformation $f : R \to R'$ must satisfy such that Laplace's equation remains invariant under this transformation. For simplicity, let's consider the simplest case in $\mathbb{R}^2$.

Applying the chain rule we have that:

$$\begin{aligned}\frac{\partial\phi}{\partial x} &= \frac{\partial\phi}{\partial x'}\frac{\partial x'}{\partial x} + \frac{\partial\phi}{\partial y'}\frac{\partial y'}{\partial x}\\ \frac{\partial\phi}{\partial y} &= \frac{\partial\phi}{\partial x'}\frac{\partial x'}{\partial y} + \frac{\partial\phi}{\partial y'}\frac{\partial y'}{\partial y}\end{aligned} \tag{14.2}$$

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